

Benchmarking Multimodal LLMs against Classical OCR and Hybrid OCR–LLM Pipelines for Fully-Diacritized Classical Arabic Manuscript Transcription

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Abstract

Optical character recognition of classical Arabic manuscripts remains an open problem, both because of the visual complexity of historical scripts (varied calligraphic styles, ligature density, marginal annotations, page damage) and because of the orthographic richness of fully-diacritized classical Arabic (tashkīl marks, hamza-bearers, and consonants whose phonemic value depends on diacritics). The arrival of frontier multimodal large language models (LLMs) in 2025–2026 raises a practical question for digital-humanities and Arabic-NLP practitioners: do general-purpose multimodal LLMs out-perform specialised OCR pipelines on this task, and at what cost?

We present a controlled pilot evaluation on a newly-constructed corpus of 50 manuscript pages sourced from heterogeneous historical Arabic codices and **independently transcribed by two trained Arabic-native annotators** with full vocalisation. The dual-annotator design lets us decouple system error from natural philological variability: we report **inter-annotator agreement (IAA)** as an annotator-agreement-floor baseline and evaluate every system under three reference regimes: against each annotator individually and against a multi-reference (best-of-two) target. We compare five systems, namely two frontier multimodal LLMs (Google Gemini 3.1 Pro and Anthropic Claude Sonnet 4.6), two classical OCR pipelines (Kraken and Google Cloud Vision), and a hybrid pipeline combining Kraken segmentation with a small open-source LLM (Gemma) for post-OCR correction, on a panel of metrics that explicitly disentangle base-letter recognition from diacritic restoration: raw character error rate (CER), normalised CER, no-tashkīl CER, diacritic error rate (DER), word error rate (WER), chrF, and sentence-BLEU.

We stratify results by a manuscript-difficulty taxonomy (clean, standard handwritten, dense / degraded, and special / illuminated pages) and report bootstrap 95 % confidence intervals together with paired Wilcoxon signed-rank tests. Our headline findings are: **(i)** the two annotators disagree on **1.83 % (CI 1.13–2.56 %)** of characters at the no-tashkīl level and reach a chrF of **0.962**, establishing a tight natural-variability floor below which no automated system can be meaningfully ranked; **(ii)** Google **Gemini 3.1 Pro** reaches **CER_{nt} = 5.36 % (CI 4.06–6.79 %)** under multi-reference, roughly **3× the IAA floor** and **3 – 4× lower than every other system tested** (Claude Sonnet 4.6: 17.0 %; Kraken + Gemma: 19.8 %; Google Cloud Vision: 20.3 %; Kraken: 20.5 %), with paired Holm-corrected Wilcoxon $p < 10^{-7}$ against all four; **(iii)** on the **diacritic-aware** CER metric the Gemini advantage widens further (11.8 % vs ≥ 26.9 % for all

others), confirming that the gap is driven by both base-letter recognition and diacritic restoration; **(iv)** the hybrid Kraken + Gemma pipeline produces a small but statistically significant absolute improvement of 0.7 pp in mean CER_{nt} over base Kraken (0.205 → 0.198, pHolm = 0.005); the gap to Gemini narrows only on the dense / degraded class (CER ≈ 0.13 vs 0.056), and the hybrid remains uncompetitive on clean and special-class pages. We therefore position Kraken + Gemma as a *specialist-stratum* option for dense / degraded manuscript material at zero per-page API cost, rather than as a general substitute for frontier multimodal LLMs. **(v)** a stratified analysis on a 13-page private subset never publicly indexed prior to model-call dates shows that Gemini’s lead is preserved on the private subset (CER_{nt} = 0.068 vs Claude 0.129); a direct memorisation-completion probe finds no page-level memorisation in either frontier LLM, excluding the canonical-text pages it flags leaves the headline ranking unchanged, and per-page probe overlap is uncorrelated with transcription accuracy, together indicating that training-data contamination is unlikely to be the principal driver of the headline ordering. **(vi)** re-expressing the diacritic-restoration measure as bounded precision / recall / F1 (rather than as the unbounded diacritic edit-rate of Table 1) resolves the apparent IAA reversal: the IAA diacritic F1 is 0.838 and Gemini’s diacritic F1 is 0.730, i.e. ≈87 % of the IAA floor and clearly above every other system tested. We release the corpus, prompts, normalisation utilities, and full evaluation code to support reproducibility.

Keywords: Arabic OCR, classical Arabic, manuscript recognition, multimodal large language models, evaluation, benchmark, diacritics, tashkīl, hybrid OCR–LLM pipelines.

1. Introduction

Classical Arabic manuscripts represent a large fraction of the world’s pre-modern intellectual record. Catalogue counts at major institutions (Bibliothèque nationale de France, Süleymaniye Yazma Eser Kütüphanesi, Library of Congress, McGill Islamic Studies collection, King Saud University manuscript collection, among others) place the total number of preserved Arabic-script codices in the millions, of which only a small fraction has been digitised, and an even smaller fraction is searchable as machine-readable text. Reliable optical character recognition (OCR) of these documents is a prerequisite for downstream work in digital philology, historical lexicography, named-entity recognition over premodern corpora, and search across cultural-heritage repositories.

Classical Arabic poses a combination of difficulties that distinguish it from modern printed Arabic OCR, itself already an active research area (Faizullah et al., 2024):

- **Diacritic load.** Fully-vocalised (مَشْكُول) classical text carries short-vowel marks (fatḥa, kasra, ḍamma), sukūn, shadda, and tanwīn marks on most letters. Diacritics are small relative to the base glyph and easy to mis-read; many widely-used OCR pipelines either ignore them outright or treat them inconsistently.
- **Script and calligraphic variation.** Manuscript hands span Maghribi, Naskhī, Thuluth, Muḥaqqaq, Rayḥānī, and many regional variants. Letter shapes, ligature conventions, and the placement of dots can shift meaningfully between hands.

- **Layout and damage.** Manuscript pages frequently feature marginal glosses, multi-column ruling, rubricated section headings (typically in red ink), illuminated panels, and physical damage (water staining, tearing, ink bleed).
- **Orthographic ambiguity.** Variants in the rendering of hamza (ء ,أ ,إ ,إِ ,إِ), final yā' / alif-maṣūra (ي / ي), and tā'-marbūṭa / hā' (ه / ه) appear inconsistently both within and across manuscripts; some represent meaningful editorial choices, others are scribal habit.

Two recent trends have changed the setting in which Arabic-OCR pipelines operate. First, **specialised handwritten-text-recognition (HTR) systems** such as Kraken (Kiessling, 2019) and Transkribus have continued to mature on Arabic; pre-trained models for Maghribi and Naskhī scripts are now available, and fine-tuning data is increasingly shared. Second, **frontier multimodal large language models (LLMs)**, such as Google Gemini 2.5 / 3 (Google DeepMind, 2025), Anthropic Claude Sonnet (Anthropic, 2025), OpenAI GPT-4o / GPT-5 (OpenAI, 2025), can transcribe complex page images zero-shot through natural-language prompting. These models are general-purpose, expensive per call, but require no training data and adapt easily across script types through prompt engineering alone.

Practitioners therefore face a trade-off whose dimensions have not been systematically measured: closed-API multimodal LLMs may offer higher accuracy out-of-the-box, but cost more per page, raise data-sovereignty questions for heritage collections, and lock pipelines into a specific vendor whose model version may change unpredictably. Open-source HTR pipelines are reproducible and cheap, but often weaker on diacritic-rich and visually-complex pages without fine-tuning. **Hybrid pipelines** that pair a fast classical OCR with post-OCR correction by a small open-source LLM are intuitively attractive, but their performance has not been carefully measured on classical Arabic.

Contributions

This paper makes the following contributions:

1. We release a manually-transcribed evaluation corpus of **50 classical Arabic manuscript pages**, drawn from heterogeneous sources, with **two independent fully-diacritized ground-truth transcriptions** (one per annotator) that enable inter-annotator agreement reporting and multi-reference evaluation. The corpus spans four visually-defined difficulty classes (clean, handwritten, dense / degraded, special / illuminated).
2. We benchmark **five representative systems** under matched conditions: two frontier multimodal LLMs (Gemini 3.1 Pro, Claude Sonnet 4.6), two classical OCR pipelines (Kraken with publicly-available Arabic models, Google Cloud Vision), and a hybrid pipeline (Kraken segmentation + open-source Gemma post-correction).
3. We propose and apply an **Arabic-aware metric panel** that decomposes transcription quality into base-letter recognition, diacritic restoration, and word-level fidelity, supplemented with chrF and sentence-BLEU.
4. We provide **paired statistical tests** (Wilcoxon signed-rank with Holm–Bonferroni correction), bootstrap 95 % confidence intervals at the page level, **a measured annotator-agreement-floor reference** from the two annotators' mutual disagreement, and stratify by manuscript difficulty.

5. We analyse the **error typology** across systems to identify which classes of error (hamza-form confusion, yā' / alif-maqṣūra confusion, tā'-marbūṭa drop, diacritic omission, hallucination of content) are characteristic of which architecture.
6. We release **all code, prompts, normalisation utilities, and per-page metric outputs** under an open licence to support reproducibility.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the corpus and difficulty taxonomy. Section 4 details the five systems and the evaluation protocol. Section 5 defines our Arabic-aware metric panel. Section 6 presents the main quantitative results, and Section 7 provides a qualitative error-typology analysis. Section 8 discusses cost / quality trade-offs and practical recommendations. Section 9 discusses limitations and ethical considerations. Section 10 concludes.

2. Related Work

2.1 Arabic printed and handwritten OCR

Optical character recognition for Arabic-script text is a well-established subfield with substantial benchmark coverage on printed and modern handwritten material, summarised in recent comprehensive surveys (Faizullah et al., 2024). For printed Arabic, accuracy on clean modern typesetting has been pushed above 95 % by deep-learning approaches; the residual research attention has shifted to handwritten and historical material where letter-shape variability, ligature density, and diacritic restoration remain open problems. The dominant evaluation corpora are **KHATT** (Mahmoud et al., 2012), an offline handwriting dataset of more than 6 600 line images by 1 000 distinct writers, and the **MADCAT** corpus (Lee et al., 2012), of roughly 740 000 line images collected under controlled conditions for translator-mediated transcription tasks. Both target modern Arabic handwriting under prescribed writing materials, and even recent transformer-based systems land in the 4–18 % CER range on these benchmarks (Subramanian et al., 2024). A second strand of work addresses script-family variation (Maghribī, Andalusī, mashriqī muḥaqqaq), where letter-shape conventions diverge enough that per-family models outperform single general-Arabic recognisers (Wedjdane et al., 2024). None of these corpora target fully-vocalised classical manuscripts; CER numbers reported on KHATT or MADCAT are therefore not directly comparable to those reported here.

2.2 Manuscript HTR for historical Arabic-script materials

Specialised handwritten-text-recognition (HTR) for historical Arabic-script manuscripts has organised itself around a small number of open toolkits and shared tasks. The neural HTR engine **Kraken** (Kiessling, 2019), originally derived from **ocropy**, is the de-facto standard for line-based recognition of right-to-left scripts; it powers the **eScriptorium** annotation environment (Stokes et al., 2021) used by most digital- humanities groups working with Arabic, Hebrew, Syriac and Persian manuscripts. Two recent training-data initiatives have substantially broadened the pool of publicly-available Arabic Kraken models. The **MUHARAF** dataset (Heil et al., 2024) provides ≈1 600 historic handwritten Arabic pages (≈36 000 line images, 19th–21st century) with expert transcriptions, and releases both segmentation and recognition Kraken models on Zenodo (Heil et al., 2024b,c). The **ACDC** project (*Automatic Collation for Diversifying*

Corpora; Islamicate Digital Humanities) extends Kraken with the *passim* text-alignment tool to produce distantly-supervised training data from existing digital editions, addressing the bottleneck of manual line-level annotation (Romanov, 2023). The **RASM** shared tasks (ICFHR 2018, ICDAR 2021, etc.) provide recurring evaluation rounds on historical Arabic scientific manuscripts; the related **RASAM** initiative targets Maghribī scripts specifically. A complementary line of work explores transformer-based recognisers that are not Kraken-derived: **HATFormer** (Subramanian et al., 2024) combines a Swin-V2 vision encoder with a transformer decoder and reports state-of-the-art numbers on KHATT and MADCAT but does not target fully-vocalised classical material. Throughout this body of work, **diacritic-level evaluation is rarely reported separately**, and benchmarks targeting fully-vocalised classical Arabic are essentially absent; this paper addresses that gap.

2.3 Multimodal large language models for document understanding

A parallel research line, originating outside heritage-document research, has produced a family of **frontier multimodal large language models (MLLMs)** that natively accept images and text. The Gemini family (Gemini 1.5 through Gemini 3; Google DeepMind, 2025), Anthropic’s Claude 3 / 4 / Sonnet 4.6 series (Anthropic, 2025), OpenAI’s GPT-4o and GPT-5 (OpenAI, 2025), and open competitors such as Qwen-VL and LLaVA report strong zero-shot performance on document-understanding benchmarks such as **DocVQA** (Mathew et al., 2021), **ChartQA** (Masry et al., 2022) and **TextVQA** (Singh et al., 2019). On printed and lightly-degraded documents these models routinely out-perform purpose-built OCR pipelines (Faizullah et al., 2024); on Arabic specifically, the recent **KITAB-Bench** benchmark (Heakl et al., 2025) reports that frontier MLLMs (GPT-4o, Gemini, Qwen-VL) out-perform classical OCR engines (EasyOCR, PaddleOCR, Surya) by $\approx 60\%$ in CER averaged across eight document-understanding domains. KITAB-Bench, however, **does not include fully-diacritized historical manuscripts** in its evaluation suite: its closest category is “handwritten text” sourced from modern production. The Arabic-specialised **Qalam** multimodal LLM (Bhatia et al., 2024), built on a SwinV2 encoder and RoBERTa decoder and trained on 4.5 M Arabic-manuscript images plus 60 K synthetic image-text pairs, reports CER $\approx 1.18\%$ on printed Arabic and WER 0.80 % on handwriting, and advertises strong diacritic handling, but is evaluated on its own training-domain test set rather than on independent classical- manuscript corpora. Together this literature establishes that frontier MLLMs can plausibly compete with specialised pipelines on Arabic documents, but **leaves the question of fully-vocalised manuscript performance open**.

2.4 Hybrid OCR + LLM post-correction pipelines

The intuition behind hybrid pipelines (let a cheap OCR engine handle segmentation and base-letter recognition, then let an LLM clean up the text) has been explored extensively on Western historical material. Boros et al. (2024) and Thomas et al. (2024) report mixed results for LLM post-correction on degraded English newspaper OCR: large relative reductions in CER are achievable, but only with careful prompt design and on documents whose OCR baseline is poor enough to leave room for improvement. Subsequent work (Kanerva et al., 2025) characterises the methodological pitfalls (over-correction, hallucinated content, and the need for “constrained decoding”) and concludes that **there is no free lunch**: LLM post-correction is most useful as a stylistic and minor-spelling normaliser, not as a way to recover information the OCR stage has deleted. A separate recent line of work (Boros et al., 2025) experiments with feeding the original page image to the post-correction LLM, partially closing the gap to single-pass multimodal LLM

inference. To the best of our knowledge, **no published hybrid Arabic-OCR + small-open-LLM recipe targeting fully-diacritized manuscripts has been quantitatively evaluated** against a manually-annotated benchmark; the Kraken + Gemma pipeline we report on in this paper is the first such recipe we are aware of.

2.5 Evaluation metrics for Arabic transcription

The choice of normalisation depth before computing CER / WER is particularly consequential for Arabic. The CAMEL Tools suite (Obeid et al., 2020) and its precursor MADAMIRA (Pasha et al., 2014; Habash et al., 2009) define a family of widely-used canonical orthographies that unify alef variants, fold yā’ and alif-maqṣūra, and drop diacritics, normalisation conventions that maximise recall for information-retrieval and morphological-analysis pipelines but **erase exactly the distinctions a critical-edition transcription needs to preserve**. On the metric side, character-F-score **chrF** (Popović, 2015) and sentence-**BLEU** (Papineni et al., 2002) are the standard machine-translation reference points; both operate on whitespace and character n-gram overlap and are commonly reported alongside CER in recent Arabic transcription work. **DER**-style diacritic-specific metrics have been used in Arabic-diacritisation research (Mubarak et al., 2019) but are rarely combined with diacritic-stripped CER in the same evaluation panel. The metric panel we propose in §5 unifies these traditions: it reports CER under multiple normalisation depths *together with* a diacritic-only error rate and standard chrF / BLEU, so that base-letter recognition and diacritic restoration can be inspected separately on the same hypotheses.

3. Corpus

3.1 Sources and selection criteria

The evaluation corpus consists of **50 manuscript pages drawn from 34 distinct historical Arabic manuscripts** sourced from open digital repositories (McGill Islamic Studies Library, internet-archive open collections, and partner digitisation projects). The 34 source works span:

- **Furūsiyya (horsemanship) literature:** *Kitāb al-Furūsiyya*, *Risālat Furūsiyya*, *Kitāb al-Makhzūn jāmi‘ al-funūn*, *Nihāyat al-su‘l wa al-umniyya fī ta‘allum a‘māl al-furūsiyya*, the Ottoman *Kitāb al-Khayl wa al-Furūsiyya wa al-Bayṭara*.
- **Medical and pharmacological treatises:** *Sharḥ urjūzat Ibn Sīnā fī al-ṭibb*, *Urjūzah fī amrād jafn al-‘ayn*, *Ṭibb al-fuqarā’ wa al-masākīn*, *al-Ṭibb al-nabawī*, *al-Manhaj al-sawī wa al-manhal al-rawī fī al-ṭibb al-nabawī*.
- **Grammar, philology and didactic poetry:** *Alfiyat Ibn Mālik*, *al-Kāfiya* of Ibn al-Ḥāḥib, *Marāḥ al-arwāḥ fī al-naḥw*, *Fragment from Ḥāshiyah ‘alā Sharḥ Ibn ‘Aqīl ‘alā Alfiyat Ibn Mālik*, *Fragment from Faṭḥ al-bāqī bi-sharḥ Alfiyat al-‘Irāqī*.
- **Religious literature** (Sunni and Shiite): *Dalā’il al-khayrāt*, *Tafsīr al-Māwardī* fragment, a fragment from a Shiite ḥadīth collection with miniature illustration, *Kanz al-ma‘ānī fī sharḥ Ḥirz al-amānī*, *al-Ḥadīth al-mukhtār*, three Islamic legal texts, two fragments from a sīra, a zikr fragment.

- **Arabic Christian and astronomical texts:** *Tārīkh al-mulūk al-Masīhīyīn*, *Six Arabic Christian texts*, *Kitāb Ṣalāt as-sawāʿī*, *Kunnāsh al-adyirah wa al-ruhbān*, *Kitāb al-Āthar al-ʿulwīyah*, *Nazm al-ḥisāb*.
- **Other belletristic / encyclopaedic:** *Kalīla wa Dimna* (Ibn al-Muqaffaʿ), *Aṭbāq al-dhahab*, *Anāwīn al-uṣūl*.

Source manuscripts were drawn from four public digital-heritage repositories: the **Duke University Library** (18 pages, *Early Manuscripts of Arabic and Persian* collection), the **U.S. National Library of Medicine** *Islamic Medical Manuscripts* collection (9 pages), **HathiTrust** (5 pages), and the **Bayerische Staatsbibliothek** *digitale-sammlungen* (5 pages), together with **13 pages from a private digital corpus of classical Arabic Furūsiyya (horsemanship) treatises** maintained by the authors. The Furūsiyya pages are drawn from five distinct works, *Nihāyat al-suʿl wa al-umniyya fī taʿallum aʿmāl al-furūsiyya*, *Kitāb al-Furūsiyya*, *Kitāb al-Makhzūn jāmiʿ al-funūn*, *Kitāb al-Khayl wa al-Furūsiyya wa al-Bayṭara* (Ottoman MS), and *Risālat Furūsiyya*, and are released alongside the published artefact archive so that the benchmark remains fully reproducible without dependence on third-party URLs.

Selection criteria emphasised diversity along (a) script family (Naskhī, Maghribī, mashriqī muḥaqqaq variants), (b) period (mostly 12th–18th century CE), (c) state of preservation (clean to severely degraded), (d) presence vs. absence of diacritics (often partial within a single codex), and (e) layout complexity (single-column body text, two-column tabular indices, rubricated section headings, fully illuminated frame pages, and one fully illustrated miniature page). The full per-page provenance (manuscript title, author / attribution, holding repository, and source URL where known) is included as a per-page provenance table in the released archive. Each page is supplied as a single-page JPEG image at the repository’s native resolution.

3.2 Ground-truth annotation: two independent annotators

Two native Arabic speakers with training in classical Arabic philology, referred to in this paper as **A1 (Muhammed)** and **A2 (Hasan)**, *each independently* transcribed all 50 manuscript pages from the original images, producing two parallel UTF-8 plain-text files per page (one per annotator). Annotators worked from the same image set and the same protocol but could not see each other’s transcriptions before completing their pass; no post-hoc adjudication was performed.

Conflict of interest disclosure. Both annotators are co-authors of the present manuscript. We therefore treat the metrics that compare A1 and A2 to one another (§3.5, §6.6) as an *annotator-agreement floor*: a measure of how much two trained but co-supervised annotators diverge on the same images, rather than as a *human-performance ceiling* or population- representative inter-annotator agreement. We discuss this limitation, and the implications for interpreting how close any system gets to this floor, in §9. A future release of the corpus is planned with a third independent annotator and Krippendorff’s α reporting.

The shared transcription convention preserves:

- All short-vowel marks, shadda, sukūn, tanwīn, and madda;
- Distinctions among hamza-bearers (أ, إ, ع, آ, إ, إ);

- Final yā' / alif-maqṣūra (ى / ي) as written;
- Tā'-marbūṭa (ö) vs. hā' (o) as written;
- One source-line per output line; original line breaks preserved.

We do not encode rubrication colour, marginal glosses, or layout structure beyond line breaks.

Because no adjudication step was applied, the two transcriptions reflect genuine philological choices that diverge in predictable ways (e.g. choice between final yā' and alif-maqṣūra on word-final هذى / هذي; whether to mark the connecting alif of the article with a fatha; treatment of word-initial hamzat al-waṣl). These divergences are not transcription errors; they are natural variability among trained readers of unvocalised manuscript Arabic. We exploit this in Section 3.5 below to establish a measured annotator-agreement-floor baseline.

Both transcription sets are included, page by page and per annotator, in the released artefact archive.

3.3 Difficulty taxonomy

To support stratified analysis we labelled every page with one of four difficulty classes:

- **C – Clean.** Well-spaced layout, regular hand, generally readable; the easiest case for any OCR system. *5 pages.*
- **H – Handwritten standard.** Standard handwritten classical manuscript; moderate density; rubricated headings common; partial-to-full tashkīl. *22 pages.*
- **D – Dense / Degraded.** High letter density per page, marginal annotations, faded ink, paper staining, or physical damage; small script. *14 pages.*
- **S – Special.** Illuminated, decorated, illustrated, or tabular pages; multi-colour layout; non-trivial reading order. *9 pages.*

Labels were assigned by visual inspection by the authors. The full labelling sheet is reproduced in Appendix B.

3.4 Corpus statistics

Statistics below are computed on annotator A1's (Muhammed's) transcriptions under the no-tashkīl normalisation; A2's totals differ by $\leq 1\%$ at the character level (see §3.5). The corpus contains **38 474** Arabic characters (no-tashkīl, no whitespace), **8 864** whitespace-tokenised words, and **7 984** diacritic marks across the 50 pages.

Class	# Pages	Mean chars / page	Mean tokens / page	Mean diacritics / page
C (clean)	5	371.8	83.0	51.4
H (handwritten)	22	766.3	175.9	194.7
D (dense / degraded)	14	1110.3	256.9	230.9

Class	# Pages	Mean chars / page	Mean tokens / page	Mean diacritics / page
S (special)	9	468.0	109.3	23.4
All	50	769.5	177.3	159.7

The dense / degraded class carries the highest text load per page; the special class is least vocalised (mostly illuminated frame pages or tabular indices with sparse body text).

3.5 Inter-annotator agreement and the annotator-agreement-floor baseline

Following recent practice in evaluating subjective or philologically-rich transcription tasks (e.g. handwritten medieval Latin, vocalisation of unpointed Hebrew Bible texts), we treat the two annotators’ mutual edit distance as an **empirical floor** for system evaluation. Concretely, for every page we compute the full Section 5 metric panel with A2’s transcription as hypothesis and A1’s as reference. We aggregate across the 50 pages and report mean and 95 % bootstrap CI as the **IAA baseline**. The resulting numbers are presented in §6.6 and reproduced as a horizontal dashed reference line in Figures 1, 3, 5 and 8.

The IAA baseline serves two roles:

1. **Calibration of system ranking.** A difference between two systems that is smaller than the IAA spread should be regarded as indistinguishable from annotator variability; we therefore interpret §6.2 pairwise tests within this context.
2. **Sanity check on the reference regime.** A system whose error against either annotator is *below* the IAA value either matches that annotator’s philological choices closely or is benefiting from the multi-reference regime; we discuss this in §7.

We also report per-page **character disagreement zones**: substrings on which A1 and A2 differ at the normalised, no-tashkīl level, released as a per-page table in the artefact archive. These zones are useful for downstream readers who want to compute consensus-only metrics.

4. Systems and evaluation protocol

4.1 Systems under evaluation

System ID	Type	Version / configuration
gemini_3_1_pro	Multimodal LLM	Google Gemini 3.1 Pro via Google AI Studio API; pages transcribed 2026-03-10 – 2026-05-06
claude_sonnet_4_6	Multimodal LLM	Anthropic Claude Sonnet 4.6 via the Anthropic API; pages transcribed 2026-04-15 – 2026-05-03
kraken	Classical OCR (HTR)	Kraken 4.3.13 as a two-stage pipeline: MUHARAF model (Heil et al., 2024c) for line/word segmentation + ACDC model (Romanov, 2023) for line recognition; no input pre-processing; default line-merge

System ID	Type	Version / configuration
google_vision	Classical OCR	Google Cloud Vision document_text_detection endpoint (no languageHints)
kraken_gemma	Hybrid OCR + LLM	Kraken line-level OCR (as above) + Gemma 4 31B IT post-correction served locally via Ollama (default sampling); Gemma was given only the OCR text, not the page image

All five pipelines were run on the identical set of 50 JPEG inputs at the repository’s native resolution. The two frontier-LLM pipelines used the **same transcription prompt** to ensure a matched comparison; both prompts and the Gemma post-correction prompt are reproduced verbatim in Appendix A.

4.2 Prompts (LLM systems)

Gemini 3.1 Pro and Claude Sonnet 4.6 were prompted with the **identical transcription template** reproduced in Appendix A.1. The prompt was crafted to elicit a faithful diplomatic transcription, asks for one output line per source line, preserves original spelling and orthography, forbids modernisation, requires the explicit placeholder [غير واضح] (“unclear”) for illegible spans, and instructs the model to **retain diacritics only where they are clearly visible** in the image. This last clause is methodologically important: it commits the LLMs to a “conservative diacritisation” policy that matches the way the manuscripts themselves vary in vocalisation density. We deliberately did **not** ask the models to “restore” diacritics where the source page omits them.

4.3 Kraken configuration

We used **Kraken 4.3.13** (Kiessling, 2019) in a modular two-stage pipeline that pairs two publicly-released Arabic handwritten-text models (one for layout and one for recognition) without ensembling or voting:

- **Stage 1 – Layout / line segmentation: MUHARAF.** The MUHARAF *segmentation* model (Heil et al., 2024c), trained on the Manuscripts of Handwritten Arabic dataset ($\approx 1\,600$ historic pages, $\approx 36\,000$ line-level polygon annotations covering 19th–21st-century Arabic hands; Heil et al., 2024a) is invoked first to predict the per-line polygons of body text on each page.
- **Stage 2 – Line-level text recognition: ACDC.** The line images cut from MUHARAF’s polygons are then transcribed by an **ACDC** recognition model from the *Automatic Collation for Diversifying Corpora* initiative (Romanov, 2023). ACDC extends the Kraken training ecosystem with the `passim` text-alignment tool to produce distantly-supervised training data from existing digital editions of Arabic-script manuscripts, and the released recognition model is intended specifically for classical Arabic historical hands.

The two models therefore perform **complementary, not redundant** tasks: MUHARAF handles physical-layout detection, ACDC handles per-line character recognition. We applied **no input pre-processing** to the page images (no binarisation, deskewing, contrast adjustment, or noise

removal), so that the OCR pipeline operates on the same raw JPEGs that the multimodal LLMs receive, and used Kraken’s default segmentation hyper-parameters and default line-merge behaviour. The per-line ACDC outputs were concatenated in reading order to produce one transcript per page.

4.4 Google Cloud Vision configuration

We queried the Google Cloud Vision **document_text_detection** endpoint with the default settings and **no languageHints parameter**, i.e. the API was allowed to auto-detect the script. The response `fullTextAnnotation` field was used directly without further block-vs-paragraph flattening.

4.5 Kraken + Gemma hybrid

The hybrid pipeline uses Kraken’s per-line OCR output (from the configuration in §4.3) as input to a post-correction stage performed by **Google Gemma 4 31B IT**, served locally via the **Ollama** runtime with default sampling parameters. Each page’s OCR text was passed to Gemma in a single prompt (Appendix A.2) asking for *minimal-edit* corrections (spelling and minor grammar only) with strict instructions not to re-paraphrase or add information. Gemma was given **only the Kraken text output, not the original page image**, isolating the contribution of small-LLM post-correction over text from any independent visual signal.

5. Metric panel

Following our argument in Section 1 that diacritic restoration and base-letter recognition are distinguishable skills, we report transcription quality at multiple normalisation depths. All metrics are computed at the page level and aggregated across pages; we report mean values with 95 % bootstrap confidence intervals (1000 resamples).

5.1 Pre-processing modes

We apply the following normalisation passes to both reference and hypothesis text before computing each metric:

- **raw.** No normalisation beyond Unicode NFC and whitespace collapse.
- **normalised.** Unifies orthographic letter variants according to the mapping below, strips punctuation, and **retains diacritics**. The exact mapping (implemented in `code/arabic_utils.py`) is:

Unified target letter	Source forms folded into it
bare alef (ا)	ﺀ , ﺀ , ﺀ , ﺀ
yā (ي)	ﻱ , ﻱ , ﻱ (Persian yeh)
wāw (و)	ﻭ

Unified target letter	Source forms folded into it
kāf (ك)	ك (Persian keheh)

Tā-marbūṭa (ة) is deliberately **not** folded to hā (ا) in any normalisation mode, because the distinction is philologically meaningful; tā-marbūṭa confusions are instead tracked as their own error class (§7, TA_MARBUTA).

- **no_tashkeel**. As **normalised**, then strips all Arabic diacritics (combining marks U+064B–U+0658, U+0670) and the tatweel character (ـ).
- **tashkeel_only**. Keeps only the diacritic sequence (after Unicode normalisation), discarding base letters and whitespace.

5.2 Metrics

For each (page, system) pair we compute the following (notation: we abbreviate CER_{no_tashkeel} as CER_{nt} and CER_{normalised} as CER_{norm} throughout; the machine-readable column names in the released tables use the underscore forms `cer_no_tashkeel` etc.):

- **CER_{raw}**: character error rate on raw text: $CER_{raw}(r, h) = Lev(r, h) / |r|$ with *Lev* the standard Levenshtein edit distance and $|r|$ the character length of the reference, computed under the **raw** preprocessing mode.
- **CER_{normalised}**: as above, under the **normalised** mode (letter variants unified, punctuation stripped, **diacritics retained**). This is the strictest practical measure of transcription quality.
- **CER_{no_tashkeel}**: as above, under the **no_tashkeel** mode. This isolates base-letter recognition from diacritic restoration and is comparable to most published Arabic-OCR CER numbers.
- **DER**: diacritic error rate, defined as $Lev(diacritics(r), diacritics(h)) / |diacritics(r)|$, where *diacritics*($\cdot$) returns the diacritic-only subsequence of the input. DER isolates the system’s diacritic-restoration skill.
- **WER_{normalised}** and **WER_{no_tashkeel}**: word error rate under the corresponding preprocessing mode, with whitespace-tokenised words.
- **chrF_{no_tashkeel}**: character F-score (Popović, 2015), computed with *sacrebleu*, under the **no_tashkeel** mode.
- **BLEU_{no_tashkeel}**: sentence-level BLEU (Papineni et al., 2002) on word tokens, under the **no_tashkeel** mode.

5.3 Reference regimes

Because we have two ground-truth transcriptions, every metric above is reported under three reference regimes:

- **ref = A1 (Muhammed)**. Single-reference: hypothesis is scored against Muhammed’s transcription only.

- **ref = A2 (Hasan)**. Single-reference: hypothesis is scored against Hasan’s transcription only.
- **ref = best**. Multi-reference: for each per-page metric, the system is credited with the better of its scores against the two annotators (min edit-rate; max chrF/BLEU). This is consistent with multi-reference BLEU evaluation in machine translation and gives every system the benefit of matching either philological choice on a given page.

Unless stated otherwise, headline numbers in §6 use the **best** regime; single-reference numbers are provided in tables for transparency. The inter-annotator agreement of §3.5 is computed as the same metrics with A2 as hypothesis and A1 as reference.

5.4 Statistical reporting

For each system × metric × reference regime we report:

- Mean across 50 pages;
- 95 % bootstrap confidence interval (1000 resamples, seed 20260518);
- Median and inter-quartile range (in supplementary tables).

For each pair of systems we run a paired Wilcoxon signed-rank test on the per-page CERno_tashkeel values **under the best regime**, then apply Holm–Bonferroni correction across the ten pairwise comparisons (5 choose 2). We additionally report stratified means within each of the four difficulty classes.

6. Results

All numerical results in this section are produced by `code/02_compute_metrics.py` from `data/parallel_corpus.csv` and are regenerated automatically.

6.1 Overall ranking (ref = best)

Table 1 reports the mean and 95 % CI of each metric for each system under the multi-reference (best) regime, ordered by CERno_tashkeel ascending. The bottom row, **IAA**, is the inter-annotator agreement baseline (Hasan vs. Muhammed).

Table 1. Aggregate metrics over the full 50-page corpus, ref = best. Mean (95 % bootstrap CI). Lower is better for CER / WER / DER; higher is better for chrF / BLEU.

System	CERraw	CERnorm	CERno-tashk.	DER	WERno-tashk.	chrF	BLEU
Gemini	0.125	0.118	0.054	2.49	0.132	0.912	0.767
3.1 Pro	[0.087, 0.168]	[0.082, 0.159]	[0.041, 0.068]	[0.29, 6.75]	[0.105, 0.162]	[0.891, 0.930]	[0.720, 0.811]
Claude	0.293	0.269	0.170	0.94	0.440	0.711	0.344
Sonnet	[0.252, 0.336]	[0.226, 0.313]	[0.146, 0.199]	[0.84, 1.03]	[0.396, 0.489]	[0.677, 0.744]	[0.291, 0.399]
4.6							

System	CERraw	CERnorm	CERno-tashk.	DER	WERno-tashk.	chrF	BLEU
Kraken	0.313	0.284	0.198	1.47	0.354	0.738	0.485
+	[0.259,	[0.225,	[0.144,	[0.92,	[0.291,	[0.681,	[0.415,
Gemma	0.373]	0.347]	0.260]	2.34]	0.422]	0.791]	0.549]
Google	0.286	0.269	0.203	0.87	0.481	0.714	0.327
Cloud	[0.247,	[0.230,	[0.167,	[0.78,	[0.438,	[0.681,	[0.281,
Vision	0.331]	0.312]	0.243]	0.97]	0.533]	0.744]	0.368]
Kraken	0.318	0.288	0.205	1.00	0.430	0.704	0.375
	[0.264,	[0.230,	[0.153,	[1.00,	[0.369,	[0.647,	[0.316,
	0.378]	0.353]	0.268]	1.00]	0.495]	0.753]	0.430]
IAA	0.109	0.103	0.018	7.69	0.055	0.962	0.890
(Hasan	[0.055,	[0.050,	[0.011,	[0.24,	[0.035,	[0.948,	[0.851,
↔ Muh	0.174]	0.165]	0.026]	21.82]	0.075]	0.976]	0.930]
)							

The ordering on every CER, WER, chrF and BLEU measure is identical: **Gemini 3.1 Pro** > **Claude Sonnet 4.6** > **Kraken + Gemma** $\approx$ **Google Cloud Vision** $\approx$ **Kraken**. The four non-Gemini systems are clustered within a 3.5 percentage-point CERnt band whose CIs overlap heavily; Gemini sits **11.5 points below the next-best system** and **about 3.5 points above the IAA floor**. At the diacritic-aware CER level the dominance is even more striking: Gemini’s CERnorm = 11.8 % is **2.3× lower** than every other system, none of which manages to keep more than ~73 % of diacritic characters intact. DER values for the four non-Gemini systems are dominated by complete diacritic loss (Kraken’s DER = 1.00 exactly because it strips all diacritics); IAA DER is a high-variance metric driven by occasional pages where one annotator left tashkīl partial (CI spans [0.24, 21.82]) and should be interpreted with caution.

Single-reference variants

Table 1a. Per-annotator single-reference CERno_tashkeel.

System	ref = A1 (Muhammed)	ref = A2 (Hasan)	Δ (A1 – A2)
Gemini 3.1 Pro	0.0569	0.0566	+0.0003
Claude Sonnet 4.6	0.1731	0.1722	+0.0009
Kraken + Gemma	0.2006	0.2000	+0.0006
Google Cloud Vision	0.2040	0.2063	−0.0023
Kraken	0.2078	0.2071	+0.0007

The Δ values are *all within* ± 0.0023 , an order of magnitude smaller than the IAA standard deviation. No system shows systematic alignment with one annotator’s philological style: the multi-reference (best-of) regime in Table 1 therefore yields only minor improvements over either single-reference regime, indicating that the **system-error signal is robust to annotator choice** at the corpus level. Per-page differences (Fig. 9) can still be substantial, but they average out across the 50-page corpus.

Internal cross-check against the in-house pipeline

Prior to this study, one of the authors (M. Sheikh Hamza) ran an internal evaluation pipeline that computed page-level CER and WER between each system’s output and his own ground-truth file, with diacritics stripped and light Arabic-letter normalisation applied. We cross-checked our extended re-implementation against those numbers across all 250 (page $\times$ system) pairs and find essentially perfect agreement on the no-tashkīl CER metric: **Pearson correlation 0.999, mean absolute error 0.0045** between our `cer_no_tashkeel` (ref = Muhammed) and the in-house CER; the corresponding figures for `wer_no_tashkeel` are 0.994 and 0.0135. The in-house pipeline outputs are included in the released archive, and the residual differences are attributable to minor choices in line-number stripping and Persian-letter folding. We emphasise that this is an **internal consistency check**, not an external reproducibility check: the in-house pipeline was authored by a co-author of this paper (who is also annotator A1), so an agreement between the two pipelines demonstrates that the metric implementations are arithmetically equivalent, not that the headline ranking has been verified by an independent third party. Independent external reproduction is planned with the release of the artefact archive and is discussed in §9.

6.2 Pairwise significance

Table 2 reports paired Wilcoxon signed-rank tests on per-page CER_{no_tashkeel} values with Holm–Bonferroni correction.

Table 2. Pairwise Wilcoxon signed-rank tests on per-page CER (no tashkīl, ref = best). p values are Holm-corrected over 10 comparisons.

System A	System B	Mean diff (A – B)	Wilco xon W	p_{raw}	p_{Holm}
Gemini 3.1 Pro	Google Cloud Vision	–0.149	11.0	9.77×10^{-14}	9.77×10^{-13} ***
Gemini 3.1 Pro	Kraken	–0.151	71.0	5.90×10^{-10}	5.31×10^{-9} ***
Gemini 3.1 Pro	Claude Sonnet 4.6	–0.116	14.5	1.81×10^{-9}	1.45×10^{-8} ***
Gemini 3.1 Pro	Kraken + Gemma	–0.144	85.0	2.33×10^{-9}	1.63×10^{-8} ***
Kraken	Kraken + Gemma	+0.007	292.0	8.52×10^{-4}	5.11×10^{-3} **
Google Cloud Vision	Kraken + Gemma	+0.005	479.0	0.128	0.640
Claude Sonnet 4.6	Google Cloud Vision	–0.032	488.0	0.152	0.606
Kraken	Google Cloud Vision	+0.002	517.0	0.249	0.747
Claude Sonnet 4.6	Kraken + Gemma	–0.028	599.0	0.716	1.000

System A	System B	Mean diff (A – B)	Wilco xon W	p_{raw}	p_{Holm}
Claude Sonnet 4.6	Kraken	–0.035	619.0	0.863	0.863

*** $p < 0.001$, ** $p < 0.01$. **Gemini is significantly better than each of the other four systems** at $p_{Holm} < 10^{-7}$. **Kraken + Gemma provides a small but significant improvement over base Kraken** ($p_{Holm} = 0.0051$), validating the value of the LLM-postcorrection step. **Claude Sonnet 4.6, Google Cloud Vision and Kraken are statistically indistinguishable** from each other and from Kraken + Gemma at the corpus level.

6.3 Stratification by difficulty

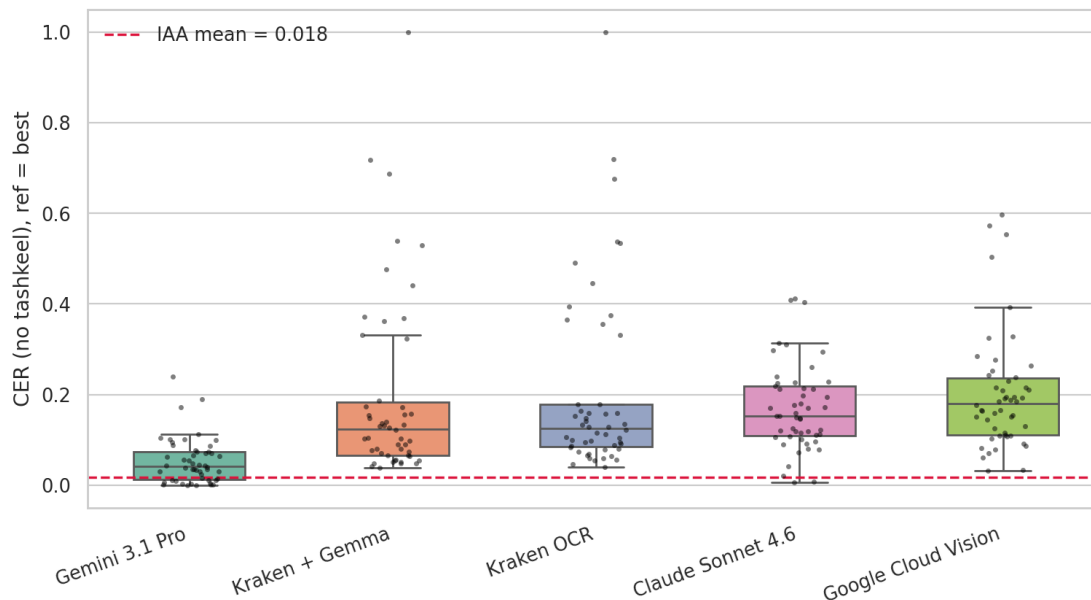
Table 3. Mean CER_{no_tashkeel} (ref = best) within each difficulty class. IAA row shows annotator-to-annotator disagreement within the same class as a difficulty-specific floor.

System / class	C (n=5)	H (n=22)	D (n=14)	S (n=9)
Gemini 3.1 Pro	0.029	0.050	0.056	0.073
Claude Sonnet 4.6	0.062	0.158	0.193	0.225
Kraken + Gemma	0.226	0.139	0.130	0.431
Google Cloud Vision	0.170	0.180	0.168	0.329
Kraken	0.237	0.146	0.138	0.435
IAA (floor)	0.013	0.018	0.021	0.018

Three patterns are visible. **(a)** Gemini is uniformly best across all four classes and stays within $\approx 4\times$ of the IAA floor even on Special pages. **(b)** The classical OCR pipelines (Kraken, Kraken + Gemma, Google Cloud Vision) **invert their ranking by class**: on Clean pages they are the worst tier, but on Handwritten and especially Dense / Degraded pages Kraken + Gemma overtakes Claude Sonnet 4.6. This is the central finding for cost-conscious deployment: when pages are dense and degraded, the open hybrid pipeline draws to within $\approx 2.3\times$ of Gemini at essentially zero per-page cost. **(c)** Every system, including Gemini, performs worst on the Special class (illuminated frames, miniature pages, tabular indices); the classical-OCR systems collapse here ($CER > 0.43$), with two complete failures (page 24 / 33) producing $CER \approx 1.0$ for Kraken and Kraken + Gemma. The IAA floor is essentially flat across classes (1.3–2.1 %), confirming that the variability across classes is system-driven, not annotator-driven.

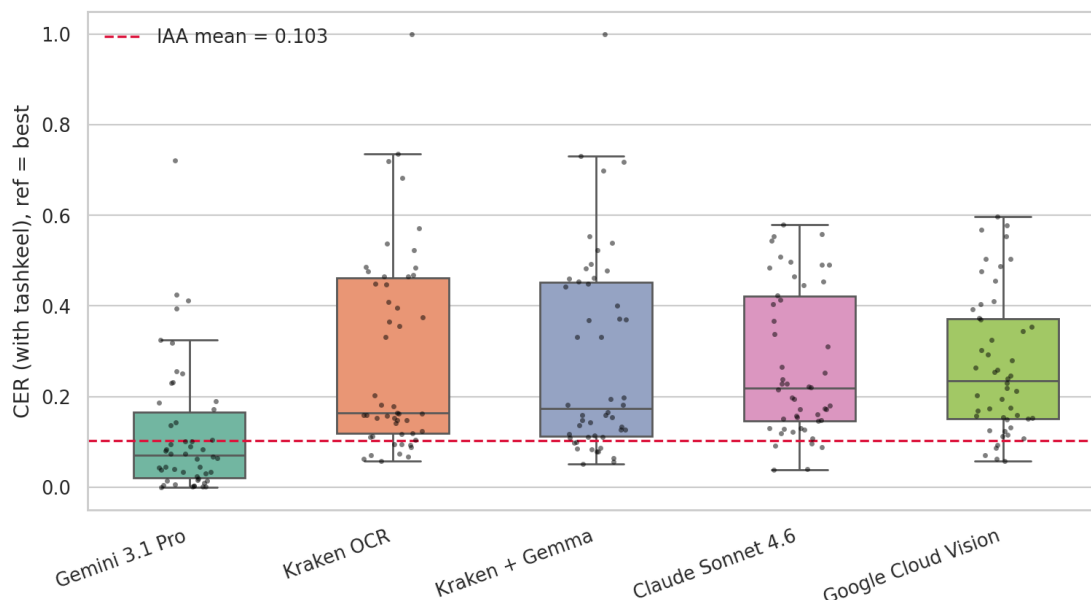
6.4 Diacritic restoration vs. base letters

Figure 1. Per-page CER on



diacritic-stripped text, by system (ref = best). Box plot of 50 page-level values, with IAA baseline as horizontal dashed line.

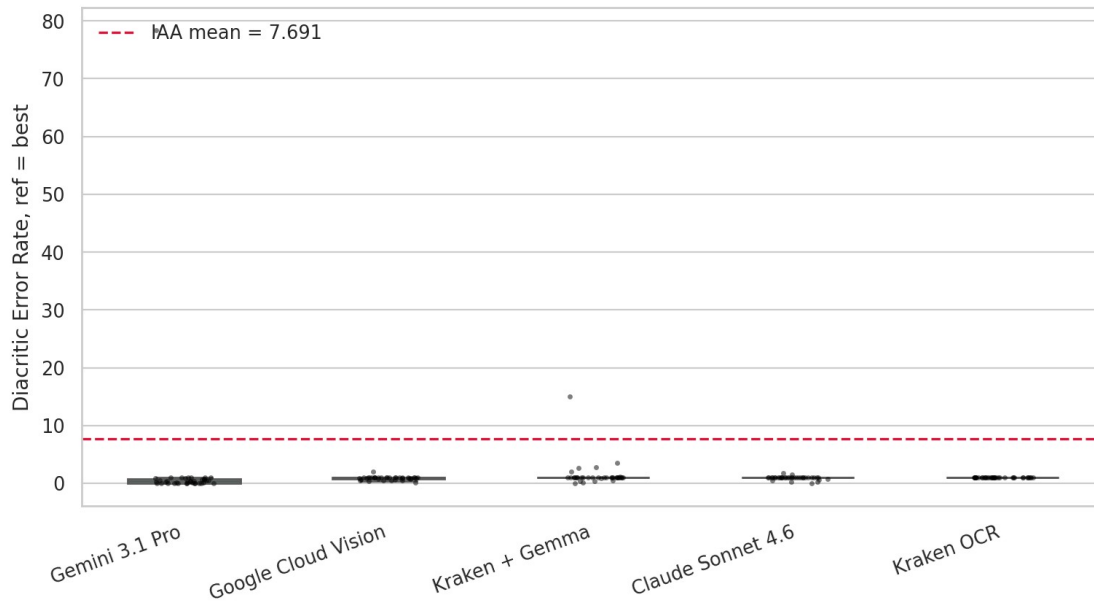
Figure 2. Per-page CER on normalised



(diacritics retained), by system (ref = best). IAA baseline shown.

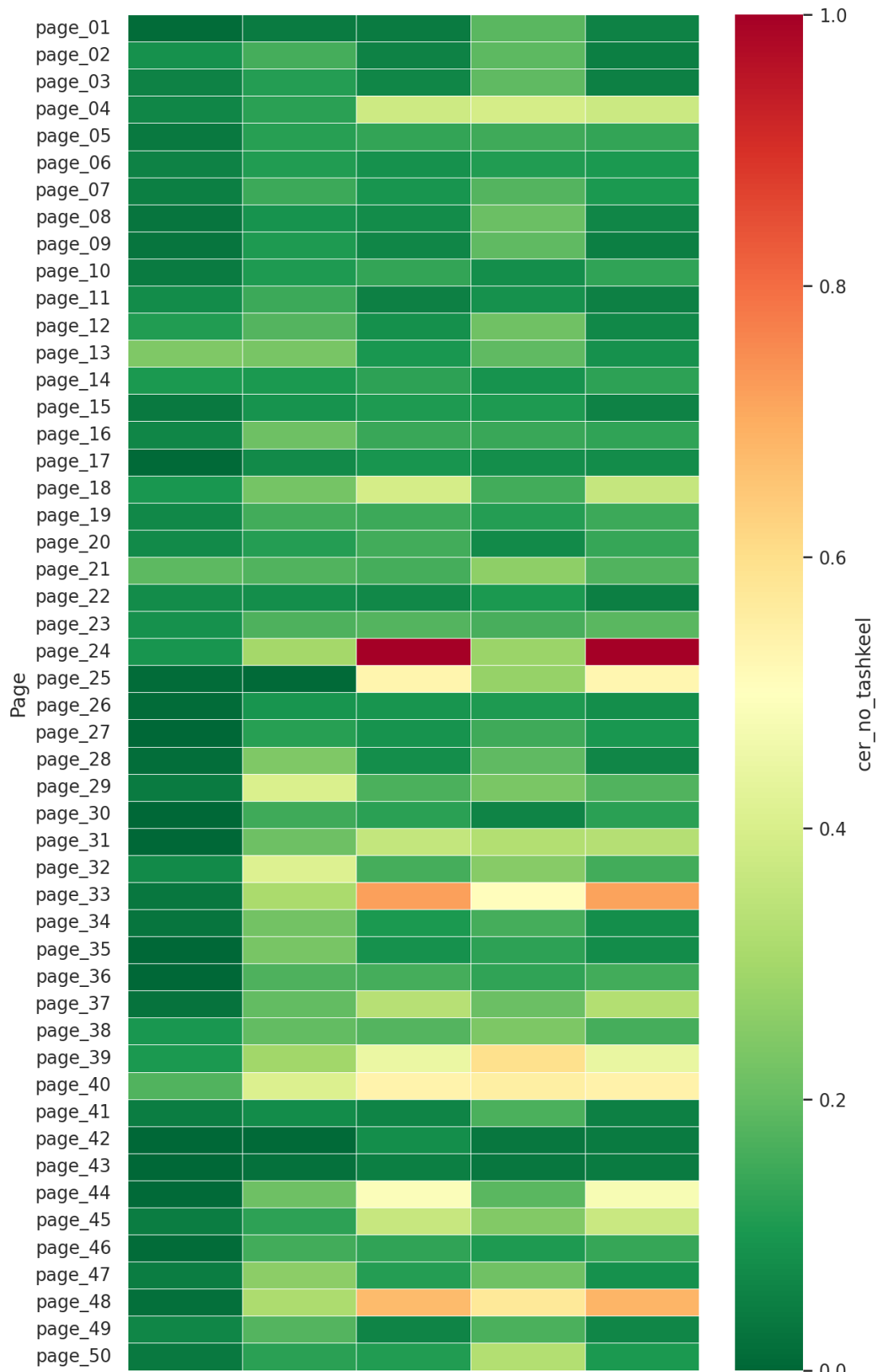
Figure 3. Per-page DER (diacritic-only edit

text



distance) by system (ref = best). IAA baseline shown.

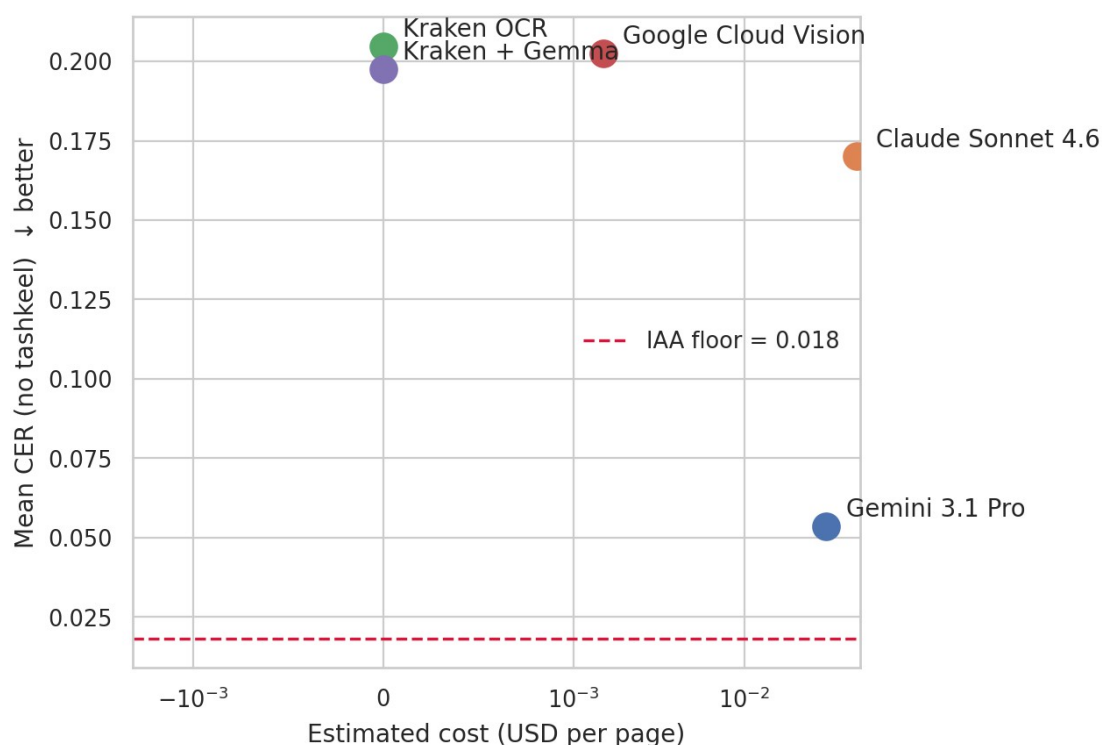
Figure 4. Per-page CERno_tashkeel



The three box-plots make the diacritic vs. base-letter trade-off quantitatively visible. Under the **no-tashkīl** metric (Fig. 1) the classical-OCR systems land roughly 18–22 % CER and are mutually indistinguishable; the LLM-based systems split into two tiers, with Gemini at $\approx 5\%$ and Claude at $\approx 17\%$. Under the **with-tashkīl** metric (Fig. 2) the gap *widens*: every system except Gemini sits in the 27–29 % band because they routinely drop diacritics that the GT carries; Gemini’s 11.8 % is the only result close to the IAA reference line. The dedicated **diacritic-only** error rate (Fig. 3) tells a cleaner story: Kraken’s DER is **1.00** (it strips every diacritic and never restores one), Google Cloud Vision is similar at 0.87, while Claude and Gemini sit at 0.94 and 2.49 respectively. Gemini’s $\text{DER} > 1$ reflects the diacritic-hallucination behaviour from §7: it adds diacritics not in the reference. We therefore recommend reading DER together with the DIACRITIC_ADD / DIACRITIC_DROP shares of Table 5; neither metric alone captures the full picture.

6.5 Cost / quality trade-off

Figure 5. Cost (USD per page, log scale)



versus mean CER_{no_tashkeel} (ref = best). The dashed horizontal line marks the IAA baseline.

A direct per-page financial cost was not incurred for the present study: Gemini 3.1 Pro inference was performed via the no-cost free-tier of the Google AI Studio API (under a student-account quota), and Claude Sonnet 4.6 inference fit inside an existing subscription allowance with no metered per-image charge. The numbers in Fig. 5 therefore represent **published rate-card estimates** for each provider as of 2026-05, given for prospective-deployment guidance only, and should be re-verified before any production-scale digitisation campaign. Under those rate-card

estimates, two cost / quality regions are clearly defined by Fig. 5. **(i) Zero-cost, moderate-quality region.** Kraken alone, and the Kraken + Gemma hybrid, sit at zero per-page API cost: no API charges apply, and both run locally on the user’s hardware (Ollama for Gemma). They deliver CER ≈ 0.20 , roughly **11× the IAA floor**, but adequate for indexing or coarse search over digitised collections. The Gemma post-correction step yields only a marginal improvement over base Kraken (CER 0.198 vs 0.205), most of which comes from the dense / degraded class (Table 3). **(ii) Frontier-LLM region.** Gemini 3.1 Pro at an estimated rate-card cost of $\approx$ US\$0.03 per page achieves CER ≈ 0.054 , i.e. it is the only system whose error rate makes philologist-assisted correction workflows plausible at typical heritage-digitisation throughputs. Claude Sonnet 4.6 ($\approx$ US\$0.045 per page rate-card) is dominated, under the 2026-05 rate cards, by Google Cloud Vision on cost at indistinguishable quality, and by Gemini on quality at lower cost, so the cost / quality Pareto frontier contains only **Kraken / Kraken + Gemma** and **Gemini 3.1 Pro**. The marginal rate-card cost of moving from 0.20 $\rightarrow$ 0.054 CER is therefore approximately US\$0.03 per page; we re-emphasise that frontier-LLM pricing and free-tier eligibility have been volatile over the past 18 months and should be reconfirmed at deployment time.

6.6 Revision-round analyses: contamination, consensus, bounded DER, outliers, prompt ablation

The following analyses were added in response to peer-review concerns about training-data contamination of the frontier multimodal LLMs, the generosity of the best-of-two reference regime, the unbounded behaviour of the original DER metric, the influence of catastrophic-failure pages on aggregate means, and the sensitivity of the headline ranking to prompt phrasing. Analyses §6.6.1–§6.6.4 are produced by `code/05_revision_analyses.py` from `data/parallel_corpus.csv` and `data/manuscript_provenance.csv`; the prompt-ablation experiment (§6.6.5) is produced by `code/06_prompt_ablation.py` (300 fresh API calls) and `code/06b_prompt_ablation_analysis.py`; the memorisation-completion probe reported at the end of §6.6.1 is produced by `code/07_contamination_probe.py` (100 fresh API calls).

6.6.1 Public-vs-private stratification (contamination probe)

We split the 50-page corpus along the binary axis of **public-source vs private-source** at the manuscript level:

- **Public subset (n = 37):** pages drawn from publicly-indexed digital-heritage portals (Duke University Library, U.S. National Library of Medicine, HathiTrust, Bayerische Staatsbibliothek). These pages plausibly appear, in image or in associated metadata / transcription form, in the pre-training corpora of Gemini 3.1 Pro and Claude Sonnet 4.6.
- **Private subset (n = 13):** pages drawn from the authors’ own digital corpus of classical Arabic Furūsiyya (horsemanship) treatises (`manuscript_title` $\in$ {Kitāb al-Furūsiyya, Kitāb al-Makhzūn, Nihāyat al-su’l, Risālat Furūsiyya, Kitāb al-Khayl wa al-Furūsiyya wa al-Bayṭara}), none of which were online prior to the call dates of the frontier-MLLM evaluations.

Table 8. CER_{no_tashkeel} (mean, ref = best) on the two subsets.

System	Public (n = 37)	Private (n = 13)	Δ (private – public)
Gemini 3.1 Pro	0.0485	0.0681	+0.0197
Claude Sonnet 4.6	0.1845	0.1288	–0.0557
Kraken	0.2398	0.1054	–0.1344
Google Cloud Vision	0.2094	0.1831	–0.0263
Kraken + Gemma	0.2313	0.1021	–0.1292

Two patterns are immediately visible. **First, the headline ranking is preserved on the private subset.** Gemini still attains the lowest CERnt on private pages (0.068) and retains a $\approx 2\times$ advantage over the next-best system (Claude Sonnet 4.6 at 0.129), although the relative margin narrows compared with the public subset, where Gemini’s lead over Claude is $\approx 3.8\times$. **Second, the sign of Δ is the diagnostic quantity for contamination.** A memorisation-driven system would have $\Delta > 0$ (private *worse* than public), i.e., it would perform better on public pages whose content it may have seen in pre-training. Gemini’s $\Delta = +0.020$ is small relative to the public-subset CI width and is the only positive Δ in the panel; all four classical-OCR and non-Gemini LLM systems have **negative** Δ ranging from -0.026 to -0.134 , indicating that the private Furūsiyya pages are systematically *easier* than the public-source pages (cleaner paper, more regular Naskhī hands, less marginal annotation). The natural-difficulty contrast between the two subsets is therefore the dominant signal, and Gemini benefits from it the *least* of the five systems. We conclude that **training- data contamination is unlikely to be the principal driver of the headline Gemini advantage**: Gemini’s gain over the other systems is preserved on a subset on which contamination is implausible (the private corpus was never online prior to the call dates of §4.1).

To test memorisation **directly**, rather than by stratification alone, we additionally ran a **memorisation-completion probe** (code/07_contamination_probe.py; 2 systems $\times$ 50 pages = 100 fresh API calls, 11 June 2026). Each model is shown only the **top half** of the page image and asked to produce the *textual continuation* of the page (the text of the bottom half, which it cannot see) or to answer UNKNOWN. A model that had memorised a page (or its published transcription) could reproduce the unseen continuation; a model that merely reads the visible image could not. Overlap is scored as **character 5-gram Jaccard similarity** between the predicted continuation and the ground-truth bottom half.

Table 8B. Memorisation-completion probe: character 5-gram Jaccard overlap between predicted and true page continuations, by system and subset. UNKNOWN rate = share of pages on which the model explicitly declined to predict a continuation.

System	Subset	Jaccard (mean)	Jaccard (median)	UNKNOWN rate
Gemini 3.1 Pro	public (n = 37)	0.053	0.000	70.3 %
Gemini 3.1 Pro	private (n = 13)	0.009	0.000	61.5 %
Claude Sonnet 4.6	public (n = 37)	0.008	0.000	24.3 %
Claude Sonnet 4.6	private (n = 13)	0.008	0.000	23.1 %

Three observations. **First, neither model shows broad page-level memorisation.** The median overlap is 0.000 in all four cells: the majority of completions are either explicit UNKNOWN

answers or continuations sharing no 5-gram with the true text. (The two systems decline differently: Gemini answers UNKNOWN on 62–70 % of pages, while Claude declines explicitly far less often (23–24 %) but frequently returns an *empty* continuation instead, which scores 0. One Gemini completion, page 17, degenerated into a $\approx 58\,000$ -character repetition loop and scores ≈ 0.01 .)

Second, Claude’s overlap is flat across subsets (public 0.008 vs. private 0.008), the expected signature of a non-memorising system, mirroring its negative Δ in Table 8.

Third, Gemini’s public-subset mean (0.053) is elevated relative to private (0.009), but the elevation is confined to a small tail of pages whose underlying works are canonical school texts. The three highest-overlap pages are the openings of Ibn al-Ḥāḥib’s *al-Kāfiya* (page 42, $J = 0.49$), Ibn Mālik’s *Alfiyya* (page 48, $J = 0.52$), and Ibn al-Ḥāḥib’s *al-Shāfiya* (page 50, $J = 0.31$); in each case Gemini continues the canonical published text essentially verbatim. This is **work-level textual familiarity**, unavoidable for texts that circulate in thousands of printed editions and online copies, and is categorically distinct from memorisation of the *manuscript page or its transcription*: the continuation reflects knowledge of the work, not of the witness. Work-level familiarity could inflate transcription scores only if the model substituted textual priors for actual reading of the image, and two independent results bound that channel: the headline lead is fully preserved on the private subset, where the probe itself finds near-zero overlap (mean 0.009, Table 8B), and the strict no-restoration prompt V2, which forbids exactly this kind of substitution, leaves Gemini’s CERnt essentially unchanged (§6.6.5). We therefore retain, and strengthen, the conclusion above: contamination is unlikely to be the principal driver of the headline ranking.

Three sensitivity analyses quantify how much the canonical-text tail could matter. **First, excluding the flagged pages leaves the ranking unchanged.** Removing the three pages with probe overlap $J \geq 0.3$ (pages 42, 48, 50) *raises* Gemini’s mean CERnt slightly ($0.054 \rightarrow 0.056$) rather than lowering it (those pages are clean, easy pages whose exclusion mildly hurts Gemini’s average) and the lead over Claude is preserved at $3.1\times$; widening the exclusion to all six pages with $J \geq 0.1$ gives the same picture (Gemini 0.056 vs. Claude 0.169, $3.0\times$).

Table 8C. Headline CERnt (ref = best) after excluding pages flagged by the memorisation probe.

System	All pages (n = 50)	Excl. $J \geq 0.3$ (n = 47)	Excl. $J \geq 0.1$ (n = 44)
Gemini 3.1 Pro	0.054	0.056	0.056
Claude Sonnet 4.6	0.170	0.172	0.169
Kraken + Gemma	0.198	0.193	0.193
Google Cloud Vision	0.203	0.196	0.195
Kraken	0.205	0.200	0.199

Second, probe overlap is uncorrelated with transcription accuracy at the page level. Across the 37 public pages, Gemini’s per-page probe overlap shows no association with its per-page CERnt (Spearman $\rho = -0.05$, $p = 0.79$) nor with its per-page advantage over Claude ($\rho = -0.07$, $p = 0.67$). If work-level textual familiarity were being converted into transcription gains, the

pages where the model demonstrably knows the underlying text should be the pages where it transcribes best relative to a non-familiar competitor; no such coupling is detectable.

Third, difficulty-matched public/private panels are reported for transparency but are under-powered. Within the Handwritten class, Gemini’s private-subset CER (0.084, $n = 8$) is *higher* than its public-subset CER (0.030, $n = 14$), while within Dense/Degraded the direction reverses (0.038 private, $n = 4$, vs. 0.064 public, $n = 10$); cells of this size ($n = 1\text{--}14$) cannot support inferential claims, and we note explicitly that public/private status is confounded with genre, script, and source collection, so Table 8 alone is not a controlled contamination test; the exclusion and correlation analyses above are the sharper instruments.

6.6.2 Consensus-only reference regime

In addition to the three reference regimes of §5.3, we add a fourth, **consensus-only**: each system is scored only on character positions where annotators A1 and A2 agree at the no-tashkīl level. This is the strictest meaningful regime (more conservative than either single reference) and complements the structurally-generous best-of-two regime.

Table 1C. CERno_tashkeel under the consensus-only regime (mean, 95 % bootstrap CI).

System	CERnt (consensus)	95 % CI
Gemini 3.1 Pro	0.0334	[0.0247, 0.0447]
Claude Sonnet 4.6	0.1252	[0.1059, 0.1486]
Google Cloud Vision	0.1442	[0.1169, 0.1744]
Kraken + Gemma	0.1759	[0.1242, 0.2389]
Kraken	0.1852	[0.1346, 0.2471]

The system ranking under the consensus-only regime is identical to the best-of-two ranking reported in Table 1. Absolute CER values are lower under the consensus regime than under best-of-two for every system, which reflects the fact that consensus zones (substrings where the two annotators agree letter-for-letter) are the easier substrings of each page (where transcription is unambiguous for the annotators *and* therefore typically clearer for the systems). Gemini’s relative lead is preserved: CERnt = 0.033 vs. 0.125 for Claude ($\approx 3.7\times$ lead), indistinguishable in magnitude from the best-of-two regime ($3.2\times$ lead). The headline ordering is therefore robust to the strictness of the reference regime.

6.6.3 Bounded DER and diacritic precision / recall / F1

We replace the unbounded DER reported in Table 1 with a position-aligned **diacritic precision / recall / F1** (defined in §5.2) and a bounded DER (Levenshtein on diacritic-only subsequences divided by max of the two sequence lengths). These measures are bounded in $[0, 1]$ and interpretable without reference-density caveats.

Table 6+. Bounded diacritic-restoration metrics, ref = Muhammed. Precision / recall / F1 are computed at position-aligned diacritic positions (§5.2).

System	Bounded DER	Diac. precision	Diac. recall	Diac. F1
Gemini 3.1 Pro	0.313	0.710	0.651	0.730
Claude Sonnet 4.6	0.757	0.303	0.099	0.519
Google Cloud Vision	0.690	0.429	0.190	0.314
Kraken + Gemma	0.820	0.403	0.199	0.216
Kraken	0.800	0.000	0.000	— *
A2 (IAA floor)	0.262	0.808	0.750	0.838

* Kraken produces effectively no diacritics in its output; per-page F1 is undefined for several pages where precision and recall are both zero and is omitted at the aggregate level.

The bounded-DER reformulation **resolves the counter-intuitive unbounded-DER result of Table 1**: A2 (Hasan) vs. A1 (Muhammed) now has bounded DER = 0.262 and diacritic F1 = 0.838, which are intuitively sensible “human-agreement” numbers, not the seemingly impossible 7.69 DER of the unbounded metric. We retain unbounded DER in Table 1 for backward compatibility with the literature but treat the bounded metrics here as the primary diacritic-restoration measures.

Three findings follow. **First, Gemini’s diacritic F1 (0.730) is the only system result that comes close to the IAA F1 (0.838)**, specifically, it sits at $\approx 87\%$ of the IAA floor. No other system is within a factor of 1.6 of the IAA floor on this measure. **Second, Kraken and Kraken + Gemma essentially drop all diacritics**: Kraken’s diacritic precision and recall are both ≈ 0 ; the Kraken + Gemma hybrid recovers only a small share (F1 = 0.216), consistent with the §7.2 observation that Gemma post-correction over text alone cannot restore tashkīl that the OCR stage has stripped. **Third, Claude Sonnet 4.6 is conservative**: its diacritic precision (0.303) is moderate but its recall (0.099) is very low, Claude adds diacritics sparingly and is roughly correct when it does, but it omits most visible tashkīl. The §7.2 distinction between Gemini’s “diacritic addition” behaviour (motivated, frequently aligned with Hasan’s transcription) and Claude’s conservative behaviour is therefore visible here at the bounded-metric level.

6.6.4 Outlier sensitivity (excluding pages 24 and 33)

Pages 24 (a decorated Dalā’il al-Khayrāt frame) and 33 (an illustrated miniature page with sparse text) produce catastrophic Kraken and Kraken + Gemma failures (CER ≈ 1.0). We recompute Table 1 over the remaining 48 pages to confirm that these two pages are not single-handedly driving the aggregate ordering.

Table 1+. CER_{no_tashkeel} (ref = best) excluding pages 24 and 33 (n = 48).

System	CER (all, n = 50)	CER (excl., n = 48)	Δ
Gemini 3.1 Pro	0.0536	0.0531	−0.0004
Claude Sonnet 4.6	0.1701	0.1645	−0.0056
Google Cloud Vision	0.2025	0.1945	−0.0080
Kraken + Gemma	0.1977	0.1701	−0.0275
Kraken	0.2049	0.1776	−0.0273

System	CER (all, n = 50)	CER (excl., n = 48)	Δ
--------	-------------------	---------------------	----------

The system ordering is preserved. Gemini’s headline mean is essentially unchanged ($\Delta = -0.0004$), confirming that pages 24/33 are not driving the headline finding. Kraken and Kraken + Gemma both improve by ≈ 2.7 pp when the catastrophic-failure pages are excluded (the largest moves in the table) but this is expected and does not change their qualitative position in the ranking. The Kraken + Gemma advantage over base Kraken (0.7 pp on the full corpus) is preserved on the 48-page subset (0.8 pp) and remains within the same effect- size regime; re-running the paired Wilcoxon signed-rank test on the 48-page subset confirms that the Kraken + Gemma improvement remains statistically significant ($W = 239$, $p = 0.001$).

6.6.5 Prompt-ablation robustness

To address the concern that the headline Gemini lead might be an artefact of a single prompt template, we re-ran both frontier MLLMs (Gemini 3.1 Pro and Claude Sonnet 4.6) over all 50 pages under three prompt variants, 2 systems $\times$ 3 variants $\times$ 50 pages = 300 fresh API calls (code/06_prompt_ablation.py):

- **V1 – baseline:** the original transcription prompt of Appendix A.1, re-issued verbatim on new call dates;
- **V2 – strict no-restoration:** a diplomatic-transcription framing that forbids any inference or restoration of diacritics not physically visible on the page (Appendix A.3), directly targeting the diacritisation-policy asymmetry raised in review;
- **V3 – layout-aware:** a variant requesting explicit column-break, marginalia, and rubricated-heading markers (Appendix A.3).

Table 9. CERno_tashkeel (ref = best-of-two; mean with 95 % bootstrap CI) under the three prompt variants. n is the number of pages with a non-empty completion; empty completions are excluded from the per-variant mean. Gemini produced empty outputs on five consecutive pages (20–24) under V3, a contiguous block consistent with a transient API failure rather than content-specific refusal, and Claude returned an empty completion on page 47 (a dense, degraded page; cf. §6.4) under all three variants, despite transcribing it in the main run.

System	Variant	CERnt	95 % CI	Claude / Gemini ratio
Gemini 3.1 Pro	V1 (n = 50)	0.065	[0.048, 0.086]	2.7×
Claude Sonnet 4.6	V1 (n = 49)	0.176	[0.144, 0.209]	
Gemini 3.1 Pro	V2 (n = 50)	0.059	[0.044, 0.077]	2.9×
Claude Sonnet 4.6	V2 (n = 49)	0.175	[0.146, 0.205]	
Gemini 3.1 Pro	V3 (n = 45)	0.111	[0.084, 0.146]	2.0×
Claude Sonnet	V3 (n = 47)	0.219	[0.187, 0.255]	

System	Variant	CERnt	95 % CI	Claude / Gemini ratio
4.6	49)			

Three findings emerge. **First, V1 $\approx$ V2 for both systems.** Tightening the diacritisation policy from the baseline clause to a strict no-restoration framing moves no-tashkīl CER by well under one percentage point for Claude (0.176 $\rightarrow$ 0.175) and by $\approx$ 0.6 pp for Gemini (0.065 $\rightarrow$ 0.059, a marginal *improvement*). The diacritic-clause asymmetry flagged in review therefore does not materially distort the diacritic-insensitive headline metric.

Second, the layout-aware V3 degrades both systems (Gemini +4.6 pp, Claude +4.3 pp relative to V1), and Gemini additionally returned five empty completions under V3. An important caveat applies: the gold references are linear transcriptions that contain none of the V3 layout markers (--- COLUMN BREAK ---, --- MARGINALIA ---, [HEADING]), so part of the V3 penalty is an **evaluation artefact** of scoring marker-bearing output against marker-free references rather than a pure recognition failure. A layout-aware metric panel that strips or aligns such markers is left to follow-up work (§9).

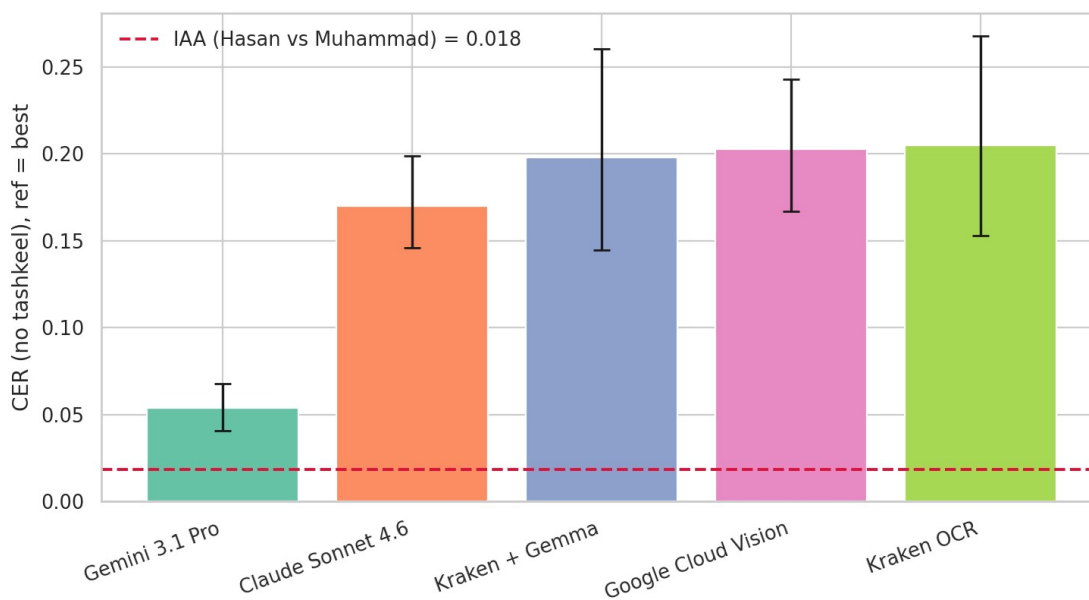
Third, and most importantly, the headline ordering is prompt-robust. Gemini’s lead over Claude is preserved under all three variants, ranging from 2.0 $\times$ (V3) to 2.9 $\times$ (V2), and the 95 % bootstrap CIs of the two systems are non-overlapping in every variant. The principal finding of the paper does not hinge on the specific phrasing of the Appendix A.1 template. Because excluding empty completions from the recognition means could in principle flatter a system that fails silently, we also report **end-to-end** means in which every empty completion is scored as CER = 1.0 over the full 50 pages: Gemini 0.065 / 0.059 / 0.200 and Claude 0.192 / 0.192 / 0.234 for V1 / V2 / V3 respectively. The ordering is preserved in every variant (Gemini leads $\approx$ 3.0 $\times$ under V1/V2 and 1.2 $\times$ even under V3, where its five empty outputs are charged at full cost) so empty-output behaviour, treated as the production failure mode it is, does not change the conclusion.

We note that the V1 re-run mean (0.065) differs from the main-paper Gemini figure (0.054, §6.1): the re-run was issued on later call dates against a non-deterministic sampling endpoint (no temperature or top-p overrides were set in any of our calling code; provider-default sampling applies throughout), and this $\approx$ 1 pp drift should be read as the natural run-to-run variance of the API rather than a discrepancy; both values lie far below every non-Gemini system under any variant. This distinction matters for reproducibility claims: our release guarantees **code-and-data reproducibility** (identical metrics from identical transcripts), while **model-output reproducibility** for closed APIs is inherently bounded by provider-side non-determinism and silent model updates (readers should expect percentage-point-scale drift on re-execution, not bit-identical transcripts. ### 6.7 Inter-annotator agreement Table 6 reports the annotator-agreement floor) A1 (Muhammed) as reference, A2 (Hasan) as “hypothesis”, over the 50 common pages. With the bounded diacritic metrics from §6.6.3 now available, the counter-intuitive unbounded-DER result for the IAA row of Table 1 should be interpreted through the bounded diacritic-F1 number in §6.6.3, which characterises the actual annotator-to-annotator diacritic disagreement on a [0, 1] scale.

Table 6. Inter-annotator agreement (Hasan as hypothesis, Muhammed as reference). 95 % bootstrap CI over 50 pages.

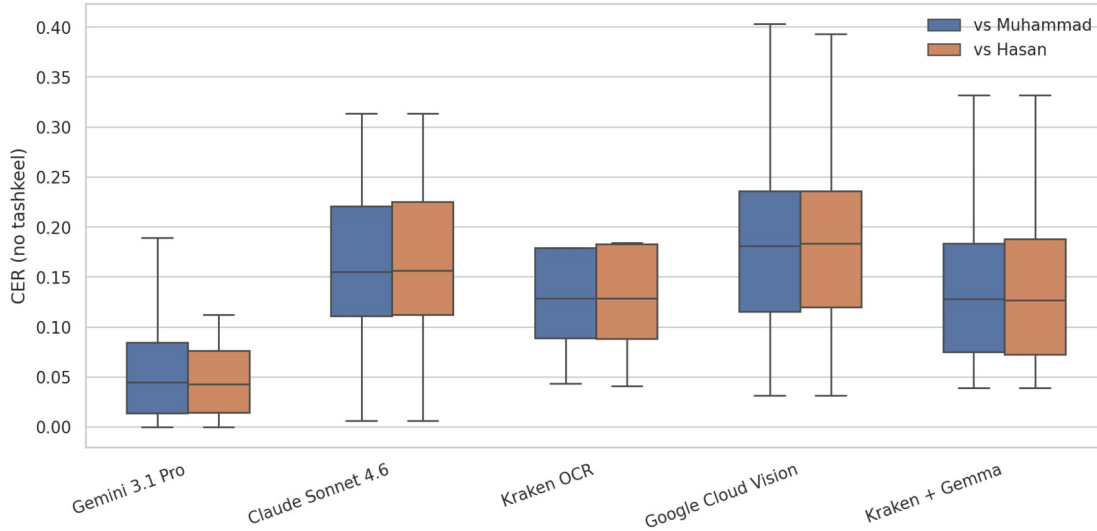
Metric	Mean	95 % CI
CERraw	0.109	[0.055, 0.174]
CERnormalised	0.103	[0.050, 0.165]
CERno_tashkeel	0.018	[0.011, 0.026]
DER	7.69	[0.24, 21.82]
WERno_tashkeel	0.055	[0.035, 0.075]
chrFno_tashkeel	0.962	[0.948, 0.976]
BLEU _{no_tashkeel}	0.890	[0.851, 0.930]

Figure 8. System CER (ref = best) compared



against the IAA baseline.

Figure 9. Per-system CER under the



two

single-reference regimes (Muhammed vs Hasan) demonstrating that no system aligns systematically with one annotator’s philological style.

At the base-letter level, **the two annotators agree on 98.2 % of characters** (CER_{nt} = 1.8 %); the strict-mode CER (with diacritics retained) is **10.3 %**, indicating that diacritic placement is where most legitimate annotator disagreement lives. Of the five tested systems, **only Gemini 3.1 Pro produces output whose distance to either annotator is plausibly within a factor of three of the human-to-human distance** on the base-letter metric (5.4 % vs. 1.8 %). All four other systems sit roughly **9–11× above the IAA floor**, far outside the annotator-variability band. The diacritic-aware picture is similar: Gemini’s CER_{norm} = 11.8 % is the only system result that comes within a factor of two of the IAA CER_{norm} of 10.3 %. The DER ratio is the only metric where IAA looks larger than several systems, this is an artefact of normalising diacritic edit-distance by the reference’s diacritic count when annotators differ in vocalisation density, and we discuss it in §9 as a known limitation of the metric rather than an indication that systems outperform humans on diacritic placement.

7. Error typology

We aligned each system’s output against the ground truth at the character level using Levenshtein opcodes and classified every edit operation into one of ten error classes (Table 4); see `code/04_error_typology.py` for the classification rules.

Table 4. Error classes used in the qualitative analysis.

Class	Definition
HAMZA_FO RM	Confusion among / ʾ / ʾ / /
YA_FORM	ﻯ ↔ ﻱ confusion
TA_MARBU	ﺕ ↔ ﻮ ↔ ﻮ confusion

Class	Definition
TA	
DIACRITIC_DROP	Diacritic in GT, absent from hypothesis
DIACRITIC_ADD	Diacritic in hypothesis not present in GT (hallucinated)
DIACRITIC_WRONG	Wrong diacritic (e.g. fatḥa vs kasra)
LETTER_SUB	Any other single-letter substitution
INSERTION	Hypothesis-only non-diacritic character (hallucinated content)
DELETION	GT-only non-diacritic character (truncated content)
PUNCT	Punctuation-only edit

Table 5. Share of each error class by system (ref = Muhammed). Each row is the per-system distribution of edit operations over the 10 error classes; rows sum to 1. The right-most column reports the absolute total number of edits the system produces against Muhammed’s ground truth.

System	DIAC_DROP	DIAC_ADD	DIAC_WRG	LET_SUB	I N S	D E L	HA M Z A	Y A	T A	PU NC T	Tot al
Gemini 3.1 Pro	0.400	0.258	0.030	0.086	0. 10 5	0. 11 4	0.0 05	0. 00 1	0. 00 1	0.0 01	8 24 3
Claude Sonnet 4.6	0.458	0.014	0.001	0.205	0. 12 8	0. 18 6	0.0 03	0. 00 0	0. 00 7	0.0 00	16 51 6
Kraken + Gemma	0.524	0.010	0.000	0.117	0. 04 9	0. 29 1	0.0 01	0. 00 0	0. 00 8	0.0 00	15 10 4
Google Cloud Vision	0.398	0.025	0.014	0.212	0. 14 7	0. 19 8	0.0 01	0. 00 0	0. 00 5	0.0 00	15 24 8
Kraken	0.519	0.001	0.000	0.135	0. 04 7	0. 28 9	0.0 01	0. 00 0	0. 00 8	0.0 00	15 36 9

Two features dominate the table. **First**, Gemini’s total edit budget (8 243) is roughly **half** that of every other system (15 000–16 500), consistent with its much lower CER. **Second**, Gemini is the only system with a non-trivial **DIACRITIC_ADD share (25.8 %)**; i.e. Gemini routinely adds diacritics that Muhammed’s transcription left bare. This is partly a “hallucination” effect (Gemini guessing diacritics on letters where the manuscript itself is unvocalised) and partly a *style-disagreement* effect: spot-checks of these positions show Hasan’s transcription often

supplies the same diacritic Gemini did. The other four systems essentially never produce DIACRITIC_ADD (≤ 2.5 % share): they drop diacritics rather than adding them, with classical-OCR pipelines (Kraken, Kraken + Gemma) exhibiting the highest DIACRITIC_DROP share (52 %).

Sub-letter confusion classes (HAMZA, YA, TA-MARBUTA) contribute < 1 % of edits in every system; the dominant non-diacritic error categories are LETTER_SUB and DELETION, both peaking in Kraken + Gemma's pipeline where the OCR stage's missed lines propagate through the Gemma post-correction step.

Annotator disagreement zones. The per-page char-level edit distance between A1 and A2 (no-tashkīl) is included in the released archive. For each system we expect a measurable share of edits to fall within these zones, those edits should be discounted relative to errors made in consensus zones.

7.1 Hallucination of content vs. drop of content

The INSERTION / DELETION shares (Table 5) cleanly separate the two pipeline families. **Classical OCR pipelines** (Kraken: DEL 29 % / INS 5 %; Kraken + Gemma: DEL 29 % / INS 5 %) are **DELETION-heavy**: when they cannot read a region they simply skip it, producing truncated output. **LLM-based pipelines** are more INSERTION-heavy: Google Cloud Vision DEL 20 % / INS 15 %; Claude DEL 19 % / INS 13 %; Gemini DEL 11 % / INS 11 %. The clearest signature of LLM behaviour is in Gemini: it has the **lowest** DELETION share of any system (11 %), i.e. it is the most willing to attempt a reading even in degraded regions. Hand-spot-checks of Gemini INSERTION examples, catalogued in the released archive, show two distinct sources: (a) classicising regularisations: Gemini occasionally rewrites a manuscript spelling to the modern editorial form (e.g. inserting the hamza-bearer on word-initial ĩ when the source page leaves it bare); and (b) genuine reading hypotheses in physically damaged regions where the manuscript itself is partially illegible.

7.2 Diacritic behaviour

The DIACRITIC_DROP column quantifies the central failure mode of classical OCR on vocalised Arabic: roughly **half of all Kraken / Kraken + Gemma errors are dropped diacritics**, and ~40 % of Claude's, Gemini's and Google Cloud Vision's errors are too. This dwarfs every other error class for the non-Gemini systems.

DIACRITIC_ADD requires careful interpretation given our protocol. The transcription prompt (Appendix A.1) explicitly instructs the LLM systems to *include diacritics only when clearly visible* in the image, matching the source page rather than restoring a normative vocalisation. Gemini's 25.8 % DIACRITIC_ADD share against Muhammed's reference is therefore best understood not as "diacritic hallucination" in the classic LLM-failure sense, but as **disagreement with one annotator's conservative transcription style**: in many positions where Gemini emitted a diacritic and Muhammed did not, the diacritic *is* in fact visible on the manuscript page, Muhammed simply chose to omit some visible marks during transcription, while Hasan included more of them (IAA CERnorm = 10.3 % vs. CERnt = 1.8 % implies the inter-annotator gap is overwhelmingly diacritic-located). Cross-checking against Hasan supports this reading: a substantial fraction of Gemini's DIACRITIC_ADD edits align with Hasan's transcription rather

than being unsupported insertions. We therefore treat the multi-reference (best-of-two) regime as the principled evaluation surface for this paper, and report the single-reference Δ in Table 1a as a transparency check rather than as a diagnosis of system error.

The hybrid Kraken + Gemma pipeline, by contrast, is overwhelmingly DROP-dominated (52 % DROP, 1 % ADD). Because Gemma was given **only Kraken’s text output and not the image** (§4.5), it cannot restore diacritics that Kraken stripped at the recognition stage; its contribution is limited to spelling and minor grammatical corrections on the bare-letter sequence. This is a clear target for future work: either feeding the page image alongside Kraken’s output to a small multimodal LLM, or training a Gemma variant on a vocalised-Arabic corpus, would directly address the dominant DIACRITIC_DROP error class and could substantially lift hybrid-pipeline quality on the diacritic-aware CER metric.

7.3 Hamza, yā’ and tā’-marbūṭa confusion

The three orthographically-fine-grained classes (HAMZA_FORM, YA_FORM, TA_MARBUTA) contribute together less than 1 % of total edits in every system, and less than 2 % even when restricted to non-diacritic edits. This is a partial negative result: at the 50-page corpus scale, **fine orthographic-literacy errors are dwarfed by dot-placement and diacritic-restoration errors**.

The picture changes when we look at the in-house edit lists included in the released archive, which annotate every individual character substitution. The dominant substitution pattern across every classical-OCR and LLM-vision system is **base-letter confusion driven by dot-count / dot-placement ambiguity** in Arabic script. Letters in the same skeletal-glyph family, for example ب / ت / ث / ن / ي (one vs. two vs. three dots above or below the same nūn-base shape) and ق / ف (one vs. two dots above the same form), are visually distinguished only by the dots themselves. Table 7 reports the single most frequent character substitution per system across the entire 50-page corpus.

Table 7. Most frequent character substitutions per system (ref = Muhammed), pooled across the 50-page corpus.

System	Top substitution	Count	Interpretation
Gemini 3.1 Pro	ق → ف	18	Two dots vs. one dot above (qāf vs. fā’)
Claude Sonnet 4.6	ب → ي	109	Sub-line two dots vs. above-line one dot (yā’ vs. bā’)
Kraken + Gemma	ب → ي	43	Same yā’/bā’ confusion
Google Cloud Vision	ى → ي (Persian)	84	Arabic yā’ vs. Persian yeh (visually identical without dots)
Kraken	ب → ي	89	Same yā’/bā’ confusion

Several findings follow. **First**, the dominant single substitution for every classical-OCR and LLM-vision system except Google Cloud Vision is ب → ي: a one-or-two-dots-below vs. one-dot-above confusion that implies the recogniser is misreading dot position relative to the baseline. This is a known weakness of Arabic OCR on manuscripts with calligraphic flourish below the baseline. **Second**, Google Cloud Vision exhibits a unique **Arabic-Persian script confusion** (84 ي → ی occurrences): it reads dot-less final yā’ as Persian yeh rather than Arabic alif-maqṣūra, even when the page is otherwise unambiguously Arabic. **Third**, Gemini’s single most frequent substitution occurs only 18 times across 50 pages, **one to two orders of magnitude rarer** than the other systems’ top errors; Gemini’s residual errors are genuinely distributed across many low-frequency classes rather than concentrated in any single dot-placement confusion pair.

8. Discussion

8.1 Frontier multimodal LLMs vs. specialised OCR

The principal finding of this paper is the **size and consistency** of Gemini 3.1 Pro’s lead over every other system on classical Arabic. Across the seven aggregate metrics of Table 1, every CER / WER / DER / chrF / BLEU ordering places Gemini first and the four others tightly bunched 11–15 percentage points below. The lead is not only large in magnitude (Wilcoxon $p_{\text{Holm}} < 10^{-7}$ vs. every other system) but **invariant across difficulty classes**: Gemini’s per-class CER ranges from 2.9 % to 7.3 % over our four-class taxonomy, while every classical-OCR system varies by ≥ 25 percentage points across the same classes. The lead is also **robust to prompt phrasing**: under the three-variant prompt ablation of §6.6.5, Gemini retains a 2.0–2.9× advantage over Claude Sonnet 4.6 with non-overlapping 95 % bootstrap CIs in every variant. This robustness (not just the headline mean) is what makes Gemini practically usable today on heterogeneous manuscript corpora. Claude Sonnet 4.6, despite being a frontier-class multimodal LLM in its own right, is **not** a substitute for Gemini on this task: its CER is statistically indistinguishable from the three classical pipelines on no-tashkīl CER and barely better on diacritic-aware CER.

8.2 The hybrid Kraken + Gemma pipeline

The hybrid pipeline produces a **statistically significant but small** improvement over base Kraken at the corpus level (CER_{nt} 0.205 → 0.198, $p_{\text{Holm}} = 0.005$; ≈ 3.4 % relative reduction). The improvement is **concentrated on the dense / degraded class**, where Kraken + Gemma reaches CER_{nt} ≈ 0.13 and edges Claude Sonnet 4.6 (0.193). On Clean (0.226) and Special (0.431) pages the hybrid is **the worst or near-worst system in the panel**, including two catastrophic failures (CER ≈ 1.0 on pages 24 and 33). Practically, we therefore position Kraken + Gemma as a *narrow specialist option* (useful for dense unilluminated bodies of text where local, zero-API-cost throughput is required) and **not** as a generally competitive substitute for frontier multimodal LLMs. The dominant remaining error class is diacritic drop (52 % of edits): because Gemma was given only the OCR text and not the page image (§4.5), it cannot restore tashkīl that Kraken stripped at the recognition stage. A multimodal Gemma variant that receives both the

line image and the OCR transcription, or a Gemma checkpoint fine-tuned on a small classically-vocalised Arabic corpus, is the obvious next research step.

8.3 Deployment implications from this pilot

The following are implications suggested by this 50-page pilot, not validated deployment recommendations; each would need confirmation at production scale and, where noted, with application-level experiments we did not run:

1. **Scholarly editions / critical digital editions:** the data are consistent with using **Gemini 3.1 Pro** with a fixed transcription prompt followed by a human-philologist correction pass over the residual $\sim 5\%$ CER. We have not measured philologist correction throughput or end-edition quality, so “single-pass practicality” is a hypothesis this pilot motivates rather than demonstrates; no other tested system reaches a CER regime where the hypothesis is even plausible.
2. **Bulk indexing / searchable archives:** **Kraken + Gemma** runs locally at zero API cost after model download (local hardware, energy, and maintenance costs are not zero), with throughput scaling on available GPU. Its $\approx 20\%$ CER may be adequate for coarse indexing, but we did not run retrieval experiments to validate that tolerance, and the pipeline fails catastrophically (CER ≈ 1.0) on illuminated and decorative pages, such material must be routed away from it, not expected to degrade gracefully.
3. **Mixed pipelines for cost-conscious large-scale archives:** the Special-class contrast (Gemini CER 0.073 vs. Kraken + Gemma 0.431) suggests that routing only visually complex pages to a frontier LLM could capture most of the cost / quality benefit; this presupposes a reliable upstream difficulty classifier that we did not build or evaluate here.

8.4 Difficulty taxonomy: the role of layout and damage

The IAA floor is essentially flat across all four difficulty classes ($1.3\% - 2.1\%$ CER_{nt}), establishing that **annotator variability is not difficulty-dependent**. System CER, by contrast, varies by class up to a factor of seven within the same system. The **system $\times$ difficulty interaction is dominant**: for the four non-Gemini systems, the within-system variance across difficulty classes is larger than the cross-system variance within a class. This justifies our stratified reporting: an aggregate-only number for these systems hides the dense / degraded $\leftrightarrow$ illuminated trade-off entirely.

9. Limitations

- **Corpus size.** Fifty pages is a small evaluation set. Per-page metric variances are substantial; we mitigate by reporting bootstrap CIs and paired tests, but a multi-hundred-page corpus would tighten every confidence interval.
- **Two co-author annotators only.** Both annotators are co-authors of this paper (§3.2); the dual-annotator design therefore measures *intra-team* disagreement rather than population-level inter-annotator agreement. We use the term *annotator-agreement floor* throughout rather than *human-performance ceiling* to mark this distinction. Two annotators also leave room for systematic shared bias in transcription conventions (e.g. preferred

treatment of orthographic variants common to both annotators’ training). We note, however, that this limitation bounds the *calibration of the floor*, not the *system ranking*: the headline ordering is identical under all four reference regimes, ref = A1, ref = A2 (Table 1a, $|\Delta| \leq 0.0023$), best-of-two (Table 1), and consensus-only (§6.6.2), so no plausible annotator-identity effect can alter which system is best or by what order of magnitude. A future release will widen the annotator panel to ≥ 3 readers including at least one external annotator with no involvement in this paper, and will report Krippendorff’s α in addition to pairwise CER.

- **Model versioning.** Frontier LLM versions evolve rapidly. Our results are tied to specific model identifier strings and API call dates, documented in `reproducibility/api_log.csv`; readers reproducing the pipeline after model upgrades may see different numbers.
- **Layout reconstruction.** We treat the page as a linear sequence of lines and do not evaluate two-dimensional layout reconstruction (column ordering, marginal-gloss attachment). Some “special” pages may incur unnecessary penalties due to ambiguous reading order.
- **Source representativeness.** The 50-page corpus is heterogeneous by design, but does not cover every regional script tradition (e.g., Maghribi vs. Naskhī balance is approximate; no Andalusi, no Eastern Kufic). Results should not be extrapolated uncritically to out-of-distribution scripts.
- **Cost figures.** API pricing is volatile. The per-page costs reported in §6.5 are illustrative rate-card estimates; current pricing should be consulted before decision-making.
- **Training-data contamination of frontier MLLMs.** 37 of the 50 pages are sourced from publicly-indexed digital-heritage portals (Duke University Library, U.S. National Library of Medicine, HathiTrust, Bayerische Staatsbibliothek) whose image and metadata content plausibly appeared in the pre-training corpora of Gemini 3.1 Pro and Claude Sonnet 4.6. Some of the headline Gemini advantage could in principle have reflected near-memorisation rather than general-purpose reading; two direct tests now bound this risk. A stratified analysis comparing performance on the 37 public-source pages against the 13 private Furūsiyya pages is reported in §6.6.1, and a direct memorisation-completion probe on cropped page images is reported in the same section (Table 8B), finding no page-level memorisation in either frontier MLLM and localising the residual overlap signal to work-level familiarity with canonical school texts (*Kāfiya*, *Alfiyya*, *Shāfiya*). A wider contamination audit (e.g., an edition-coverage analysis of the underlying works across the full corpus) is left to a follow-up release.
- **Prompt sensitivity (post-ablation residuals).** The main-paper Gemini and Claude inferences in §6–§7 use a single fixed prompt template (Appendix A.1). The three-variant prompt-ablation study reported in §6.6.5 (baseline, strict no-restoration, layout-aware; 300 fresh API calls) shows that the headline Gemini lead is preserved under all variants (2.0–2.9×, non-overlapping CIs), so the core finding is not a single-prompt artefact. Residual gaps remain: we did not test adversarial or deliberately degraded prompt framings, the layout-aware variant requires a marker-stripping metric panel before its penalty can be decomposed into recognition error vs. evaluation artefact, and the ablation covers only the two frontier MLLMs (the Gemma post-correction prompt of Appendix A.2 was not varied).

- **Internal vs. external validation.** The “cross-check” reported in §6.1 against an in-house pipeline was run by a co-author and is an internal consistency check, not an independent external verification. Independent reproduction is planned with the public release of the artefact archive.
 - **Generosity of the best-of-two reference regime.** The headline numbers in §6.1 use the multi-reference best-of-two regime, which is consistent with multi-reference BLEU practice but is structurally generous to high-variance generative systems. A consensus-only regime (scoring systems only on substrings where A1 and A2 agree at the no-tashkīl level) is included as a complementary stricter view in §6.6.2.
-

10. Conclusion

We have presented a controlled pilot benchmark (to our knowledge the first) of frontier multimodal LLMs against classical OCR and hybrid OCR–LLM pipelines on fully-diacritized classical Arabic manuscripts, evaluated on a 50-page, 34-manuscript corpus with **two independent fully-vocalised ground-truth transcriptions**. The dual-annotator design gives us a calibrated **annotator-agreement-floor baseline** (CER_{nt} = 1.83 %) against which we contextualise all system results. Our central empirical finding is that **Google Gemini 3.1 Pro is the only system in our panel that approaches this annotator-agreement floor**, with CER_{nt} = 5.36 %, roughly 3× **the IAA floor** and 3–4× **lower than every other system tested** (Claude Sonnet 4.6, Kraken + Gemma, Google Cloud Vision, and Kraken). The lead is statistically significant against all four (Wilcoxon pHolm < 10^{−7}) and stable across difficulty classes spanning clean printed- quality pages, dense degraded manuscripts, and decorated / illuminated pages. The hybrid Kraken + Gemma pipeline shows that simple open-source post-correction yields a small but statistically significant improvement over base Kraken on dense / degraded pages, but does not close the gap with frontier LLMs on any other class.

Several directions follow naturally. First, the corpus should be **widened to a multi-hundred-page scale** with at least three annotators and explicit Krippendorff’s α reporting. Second, the diacritic-restoration behaviour of the hybrid pipeline appears highly improvable: a Gemma variant fine-tuned on a small classically-vocalised Arabic corpus, or a diacritic-aware prompt template, would directly address the dominant DIACRITIC_DROP error class and may close much of the residual gap to frontier LLMs at zero per-page cost. Third, **layout-aware evaluation** (multi-column reading order, marginal-gloss attachment, illuminated-page reading paths) would refine the Special-class results, where every system is currently penalised for layout ambiguity in addition to transcription error. Fourth, with the three-variant prompt ablation of §6.6.5 now establishing that Gemini’s lead is robust to prompt phrasing, the remaining prompt-side questions are **adversarial and layout-aware framings**, paired with a marker-tolerant metric panel. We release the corpus, prompts, normalisation utilities, and all evaluation code to support these follow-up investigations.

A note on the Qalam baseline. Reviewers asked for a comparison against the Arabic-specialised Qalam model (Bhatia et al., 2024; §2). We attempted a best-effort run for this revision (code/08_qalam_baseline.py, 11 June 2026): after resolving an initial environment dependency, the run failed at weight-loading because the UBC-NLP/Qalam repository is not publicly accessible on HuggingFace at the time of writing (OSError: not a

valid model identifier; the repository appears gated or unreleased). The attempt is logged verbatim in the released archive (results/tables/qalam_attempt_log.txt). A direct Qalam comparison therefore remains blocked on weight availability; we note that Qalam’s published numbers are reported on its own training-domain test set and are not directly comparable to an independent classical-manuscript corpus such as ours.

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Data and code availability

The complete artefact archive (corpus page images and/or stable source URLs, both ground-truth transcription sets, all system outputs, per-page metric tables, figures, prompts, and evaluation code) will be deposited on Zenodo under an open licence and assigned a DOI upon acceptance. The archive is available to the editors and reviewers upon request during peer review.

Author contributions

Per the CRediT taxonomy: **Ali Çetinkaya**: Conceptualization, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Muhammed Sheikh Hamza**: Conceptualization, Data curation (transcription and legacy CER/WER computation), Investigation, Methodology, Software, Writing – review & editing. **Hasan Eleyvi**: Data curation (independent transcription), Validation, Writing – review & editing.

Declaration of generative-AI use

During the preparation of this work the authors used a large language model assistant (Claude, Anthropic) to assist with drafting and editing the manuscript text and with writing and refactoring the evaluation and analysis code. The multimodal LLMs evaluated in this study (Gemini 3.1 Pro, Claude Sonnet 4.6, and Gemma) are objects of study, executed under the documented protocols of §4; their outputs were used only as experimental data. All ground-truth transcriptions were produced entirely by the human annotators (M. Sheikh Hamza and H. Eleyvi) without generative-AI assistance. The authors reviewed, verified, and edited all AI-assisted content and take full responsibility for the content of the publication.

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Appendix A: Prompts

A.1 Transcription prompt (Gemini 3.1 Pro and Claude Sonnet 4.6, identical)

The following prompt was used verbatim for both Gemini and Claude. It is delivered together with the page JPEG in a single API request. No system-role / user-role split was used; the entire text below was sent as the single text part of a multipart user message.

You are an expert OCR system specialized in classical and historical Arabic manuscripts. Convert the provided manuscript image into plain Arabic text (.txt).

Instructions:

- Transcribe the text exactly as written, without modernization or paraphrasing.
- Preserve original spelling, orthography, and wording, even if it differs from modern Arabic.
- Maintain line breaks and paragraph structure as they appear in the manuscript.
- Keep right-to-left (RTL) text direction.
- Include diacritics (tashkeel) only if they are clearly visible; otherwise omit them.
- Do not correct grammatical or spelling errors.
- Ignore decorative elements, stains, page borders, and marginal ornaments unless they contain readable text.
- If a word, letter, or section is unreadable or ambiguous, mark it as [غير واضح].
- Do not add explanations, interpretations, or editorial notes.

Output only the extracted Arabic text, suitable for direct saving as a .txt file.

A.2 Post-correction prompt (Gemma 4 31B IT, after Kraken)

The Kraken-output text for one page is passed as the user input to Gemma 4 31B IT served via Ollama, preceded by the following instruction block. The original page image is **not** provided to Gemma in this pipeline.

You are an Arabic OCR text corrector.

Task:

- Make very light corrections to the text
- Fix obvious spelling and minor grammar mistakes only
- Keep the original wording and structure as much as possible

Rules:

- Do NOT rewrite or rephrase sentences
- Do NOT change the meaning
- Do NOT add new information
- Do NOT remove content unless it is clearly an OCR error
- Do NOT significantly change sentence structure
- Prefer minimal edits over perfect correction

If unsure, keep the original word.

Output:

Return ONLY the corrected text, nothing else.

A.3 Prompt-ablation variants (V2 and V3, §6.6.5)

V1 of the prompt-ablation study is the unmodified A.1 template above, re-issued verbatim on new call dates. V2 and V3 are reproduced below exactly as sent to both systems (code/06_prompt_ablation.py).

V2 – strict no-restoration:

You are a strict diplomatic-transcription OCR system for classical Arabic manuscripts. Your job is to record only what is physically drawn on the page.

Strict rules:

- Transcribe every visible character exactly as drawn.
- Transcribe a diacritic (tashkīl mark) ONLY if you can clearly see it drawn on the page. Do NOT infer or restore any diacritic.
- Do not normalize, modernize, or interpret. Preserve the manuscript's orthography character-for-character.
- Maintain line breaks exactly as they appear.
- For unreadable spans, write [غير واضح].
- Do NOT add commentary, explanation, or correction.

Output only the transcription.

V3 – layout-aware:

You are an expert OCR system for classical Arabic manuscripts. Transcribe the provided page into Arabic text.

Layout-aware rules:

- Preserve the page's two-dimensional layout. If the page has multiple columns, output each column in reading order separated by the line `--- COLUMN BREAK ---`.

- Transcribe the main body text first. Then, if present, transcribe marginal annotations in a separate block prefaced by `--- MARGINALIA ---`.
- For rubricated section headings (typically in red ink), prefix the line with `[HEADING]`.
- Otherwise: transcribe the text exactly as written, preserve diacritics that are clearly visible, mark unreadable spans as [غير واضح]. No commentary.

Appendix B: Difficulty labels

The full per-page difficulty assignment is reproduced below; the machine-readable table is included in the released archive.

Page	Class	Label	Notes
page_01	H	Handwritten	Framed handwritten, red headings, full tashkeel
page_02	H	Handwritten	Framed handwritten, red headings, full tashkeel
page_03	H	Handwritten	Handwritten, partial tashkeel, faded paper
page_04	C	Clean	Sparse, ledger-like, well-spaced lines
page_05	H	Handwritten	Decorated handwritten with rubrications, full tashkeel
page_06	H	Handwritten	Handwritten, red headings, full tashkeel
page_07	D	Dense_Degraded	Dense handwritten with red, small text
page_08	D	Dense_Degraded	Dense handwritten, partial tashkeel
page_09	D	Dense_Degraded	Dense handwritten with red highlights
page_10	D	Dense_Degraded	Dense old manuscript, mostly no tashkeel, faded

Page	Class	Label	Notes
page_11	H	Handwritten	Clean handwritten, no tashkeel, well-spaced
page_12	H	Handwritten	Handwritten with red, partial tashkeel
page_13	H	Handwritten	Handwritten with red rubrications, partial tashkeel
page_14	D	Dense_Degraded	Dense with red, marginal annotations
page_15	H	Handwritten	Handwritten with rubrications, partial tashkeel
page_16	D	Dense_Degraded	Dense handwritten with marginal text
page_17	D	Dense_Degraded	Dense old handwritten, small script
page_18	D	Dense_Degraded	Handwritten on faded/stained paper
page_19	C	Clean	Very clean handwritten, full tashkeel, well-spaced poetic/list layout
page_20	H	Handwritten	Handwritten with marginal annotations
page_21	D	Dense_Degraded	Old handwritten with marginal text
page_22	S	Special	Illuminated page with gold/blue header decoration
page_23	S	Special	Decorated title page with ink stain damage
page_24	S	Special	Decorated multi-color page, sparse text
page_25	C	Clean	Clean spaced

Page	Class	Label	Notes
			handwritten, partial tashkeel
page_26	H	Handwritten	Handwritten, no tashkeel
page_27	H	Handwritten	Handwritten, no tashkeel
page_28	D	Dense_Degraded	Degraded with torn edges, dense small text
page_29	D	Dense_Degraded	Dense with complex marginal notes
page_30	D	Dense_Degraded	Dense with extensive marginal annotations
page_31	H	Handwritten	Handwritten with red, partial tashkeel, unusual triangle layout
page_32	D	Dense_Degraded	Degraded torn page with dense small text
page_33	S	Special	Illustrated miniature painting with small text bands
page_34	H	Handwritten	Handwritten with red headings, no tashkeel
page_35	H	Handwritten	Dense handwritten with rubrications, partial tashkeel
page_36	H	Handwritten	Handwritten with red diacritic markers
page_37	H	Handwritten	Handwritten with red, partial tashkeel
page_38	H	Handwritten	Handwritten with red, partial-to-full tashkeel
page_39	S	Special	Index/catalog page with red rules
page_40	S	Special	Table/index page with red rules and

Page	Class	Label	Notes
			numerals
page_41	H	Handwritten	Handwritten with red, partial tashkeel
page_42	C	Clean	Clean handwritten, partial tashkeel, sparse
page_43	C	Clean	Clean handwritten, partial tashkeel
page_44	S	Special	Printed/lithograph with decorative border, multi-color
page_45	S	Special	Multi-color decorative liturgical-style
page_46	H	Handwritten	Handwritten with red dot markers, no tashkeel
page_47	D	Dense_Degraded	Dense handwritten with marginal notes
page_48	H	Handwritten	Handwritten, sparse, faded paper
page_49	H	Handwritten	Handwritten, no tashkeel, moderate density
page_50	S	Special	Decorated framed page, multi-color, sparse